

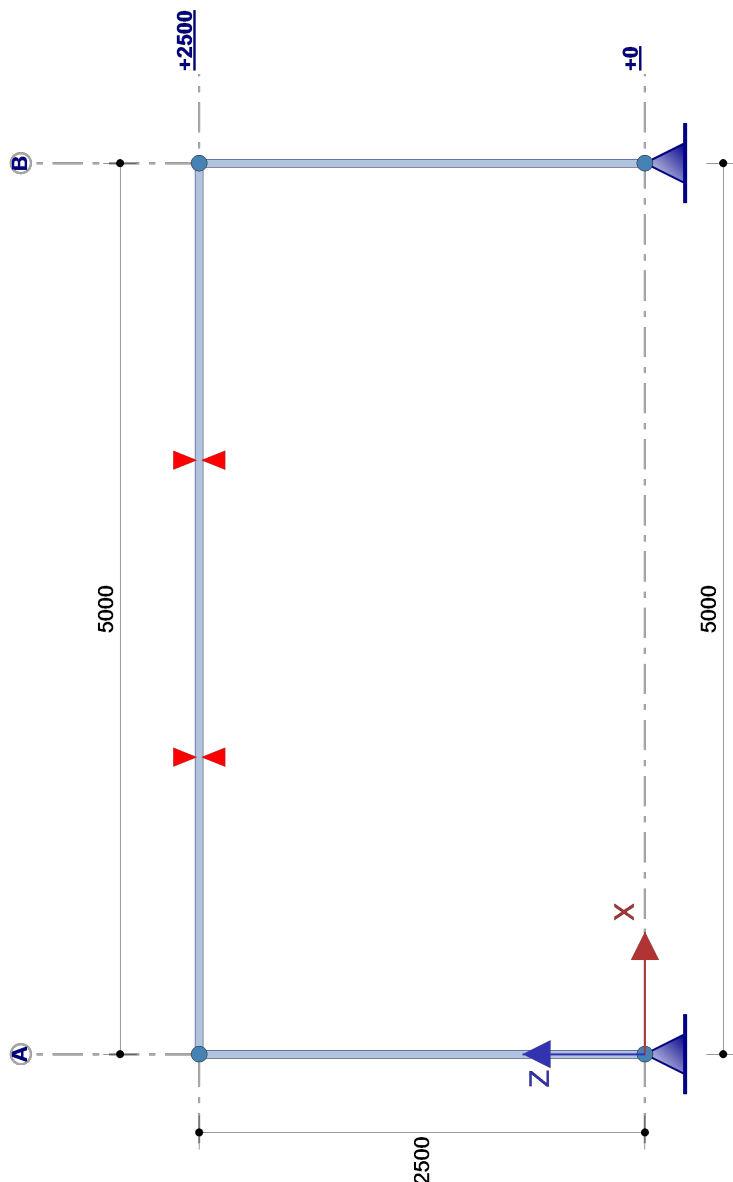
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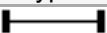
Applied standards : NEN-EN 1993-1-1+C2+A1/NB:2016 nl

Consequence class : CC1

Gravity acceleration g : 9,81 m/s²**1 Input Data****1.1 NODES**

Node Number	Coordinates		Restraints		
	X [mm]	Z [mm]	Tx	Tz	Ry
1	0	0	A	A	
2	5000	0	A	A	
3	0	2500			
4	5000	2500			

1.2 BEAMS

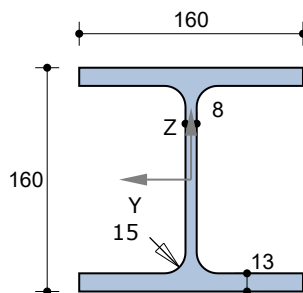
Beam Number	Node		Beam type	Profile	Length [mm]
	from	to			
1	1	3		HEB160	2500

Beam Number	Node		Beam type	Profile	Length [mm]
	from	to			
2	2	4		HEB160	2500
3	3	4		HFRHS200X200X16	5000

1.3 PROFILES

Profile Number	Name	Weight [kg/m]	E [N/mm ²]	A [mm ²]	I _y [mm ⁴]	Wy _{el_1} [mm ³]	Wy _{el_2} [mm ³]
2	HEB160	42,6	210000	5,428E3	2,4929E7	3,1161E5	3,1161E5
5	HFRHS200X200X16	90,3	210000	1,1501E4	6,393E7	6,393E5	6,393E5

HEB160



Material data

Steel grade

S 235 Hot rolled

Elasticity Modulus

E = 210000 N/mm²

Cross section data

Maximum coordinate

 $y_{max} = 80,0 \text{ mm}$ $z_{max} = 80,0 \text{ mm}$

Minimum coordinate

 $y_{min} = -80,0 \text{ mm}$ $z_{min} = -80,0 \text{ mm}$

Centroid

 $z_s = 0,0 \text{ mm}$ $y_s = 0,0 \text{ mm}$

Area / Weight

 $A = 5427,5 \text{ mm}^2$ $G = 42,6 \text{ kg/m}$

First moment of area

 $S_y = 177057 \text{ mm}^3$ $S_z = 84993 \text{ mm}^3$

Moment of inertia

 $I_y = 24929151 \text{ mm}^4$ $I_z = 8892613 \text{ mm}^4$

Radius of gyration

 $i_y = 67,8 \text{ mm}$ $i_z = 40,5 \text{ mm}$

Elastic section modulus

 $W_{y,el} = 311614 \text{ mm}^3$ $W_{z,el} = 111158 \text{ mm}^3$

Product moment of area

 $C_{yz} = 0 \text{ mm}^3$ $\text{angle} = 0,00^\circ$

Moment of inertia

 $I_{max} = 24929151 \text{ mm}^4$ $I_{min} = 8892613 \text{ mm}^4$

Radius of gyration

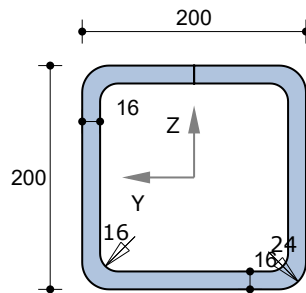
 $i_{max} = 67,8 \text{ mm}$ $i_{min} = 40,5 \text{ mm}$

Plastic neutral axis(PNA)

 $z_h = 0,0 \text{ mm}$ $y_h = 0,0 \text{ mm}$

Plastic section modulus

 $W_{y,pl} = 354113 \text{ mm}^3$ $W_{z,pl} = 169986 \text{ mm}^3$

HFRHS200X200X16**Material data**

Steel grade

S 235 Hot rolled

Elasticity Modulus

E = 210000 N/mm²**Cross section data**

Maximum coordinate

 $y_{max} =$

100,0 mm

 $z_{max} =$

100,0 mm

Minimum coordinate

 $y_{min} =$

-100,0 mm

 $z_{min} =$

-100,0 mm

Centroid

 $z_s =$

0,0 mm

 $y_s =$

0,0 mm

Area / Weight

A =

11501,3 mm²

G =

90,3 kg/m

First moment of area

 $S_y =$ 392721 mm³ $S_z =$ 392721 mm³

Moment of inertia

 $I_y =$ 63930205 mm⁴ $I_z =$ 63930205 mm⁴

Radius of gyration

 $i_y =$

74,6 mm

 $i_z =$

74,6 mm

Elastic section modulus

 $W_{y,el} =$ 639302 mm³ $W_{z,el} =$ 639302 mm³

Product moment of area

 $C_{yz} =$ 0 mm³

angle =

45,00 °

Moment of inertia

 $I_{max} =$ 63930205 mm⁴ $I_{min} =$ 63930205 mm⁴

Radius of gyration

 $i_{max} =$

74,6 mm

 $i_{min} =$

74,6 mm

Plastic neutral axis(PNA)

 $z_h =$

0,0 mm

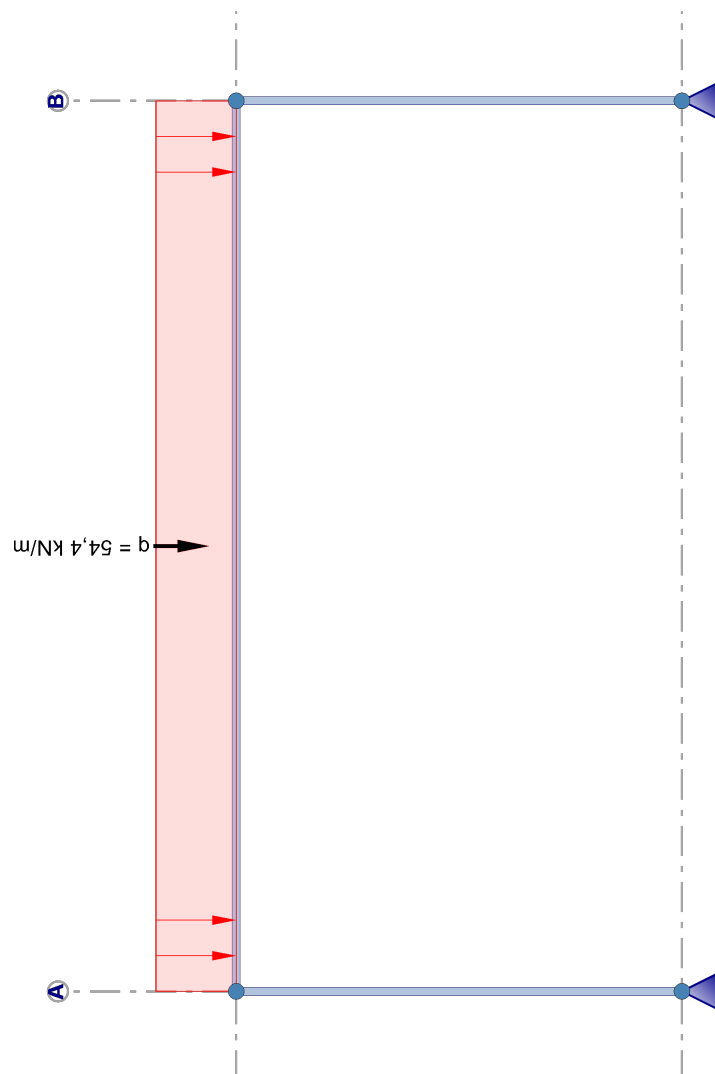
 $y_h =$

0,0 mm

Plastic section modulus

 $W_{y,pl} =$ 785442 mm³ $W_{z,pl} =$ 785442 mm³**1.4 LOAD CASES**

no.	Description	Type	ψ_0	ψ_1	ψ_2
1	Dead load	Dead load incl. self-weight	1,00	1,00	1,00
2	Live load	A:domestic	0,40	0,50	0,30

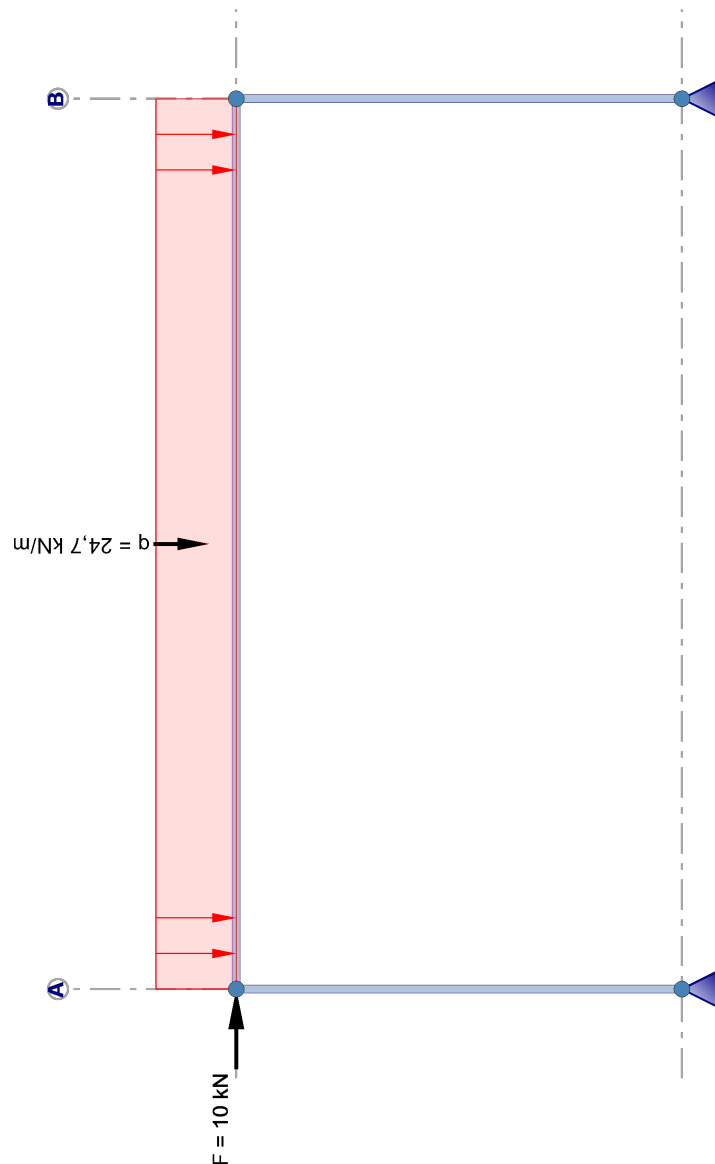
1.5 LOAD CASE 1 Dead load Including self-weight

*) Loads due to self-weight are not drawn!

Total self-weight: : 443 kg.

1.5.1 Beam loads

Beam Number	Loads				Distance from		
	Type	q1	q2	Angle	Node	a [mm]	L [mm]
3	q	-0,886 kN/m	-0,886 kN/m	0,0	3	0	5000
3	q	-54,400 kN/m	-54,400 kN/m	0,0	3	0	5000

1.6 LOAD CASE 2 Live load**1.6.1 Beam loads**

Beam Number	Loads				Distance from		
	Type	q1	q2	Angle	Node	a [mm]	L [mm]
3	q	-24,700 kN/m	-24,700 kN/m	0,0	3	0	5000

1.6.2 Node loads

Node Number	Fx [kN]	Fz [kN]	My [kNm]
3	10,000		

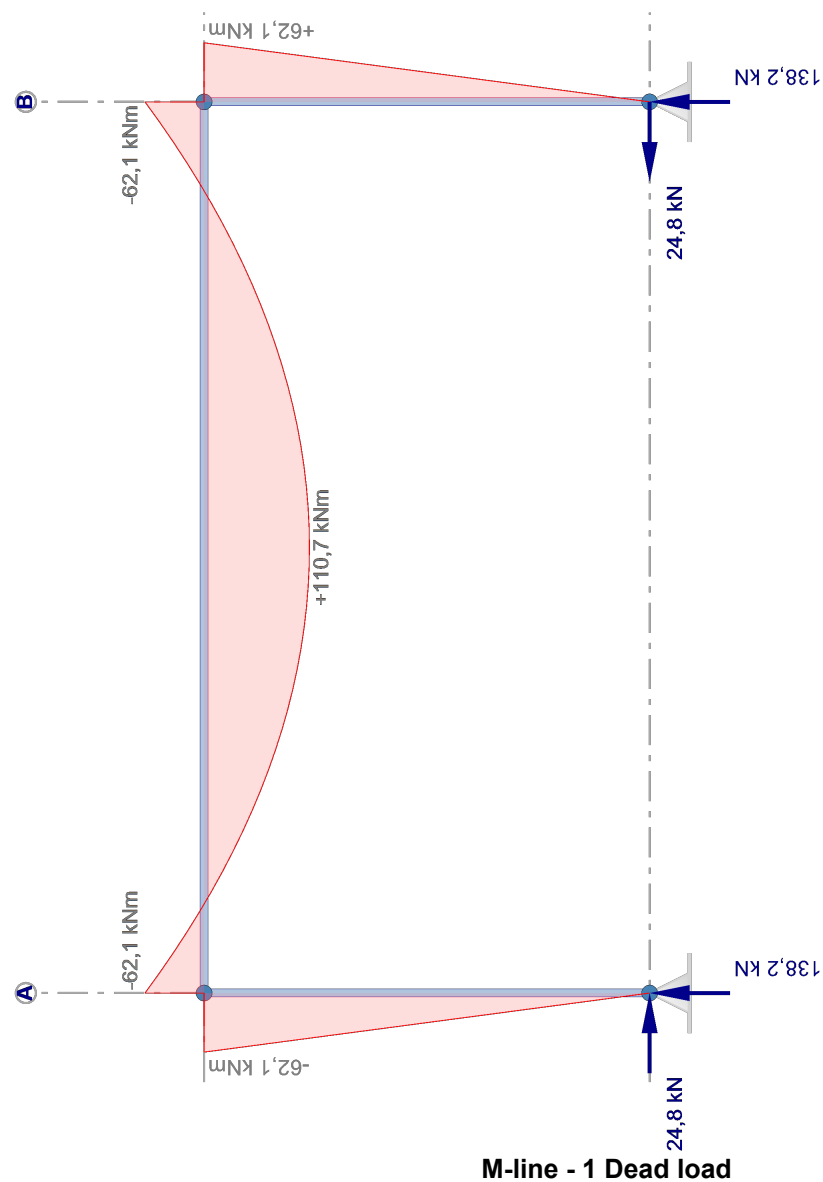
2 Calculation Results

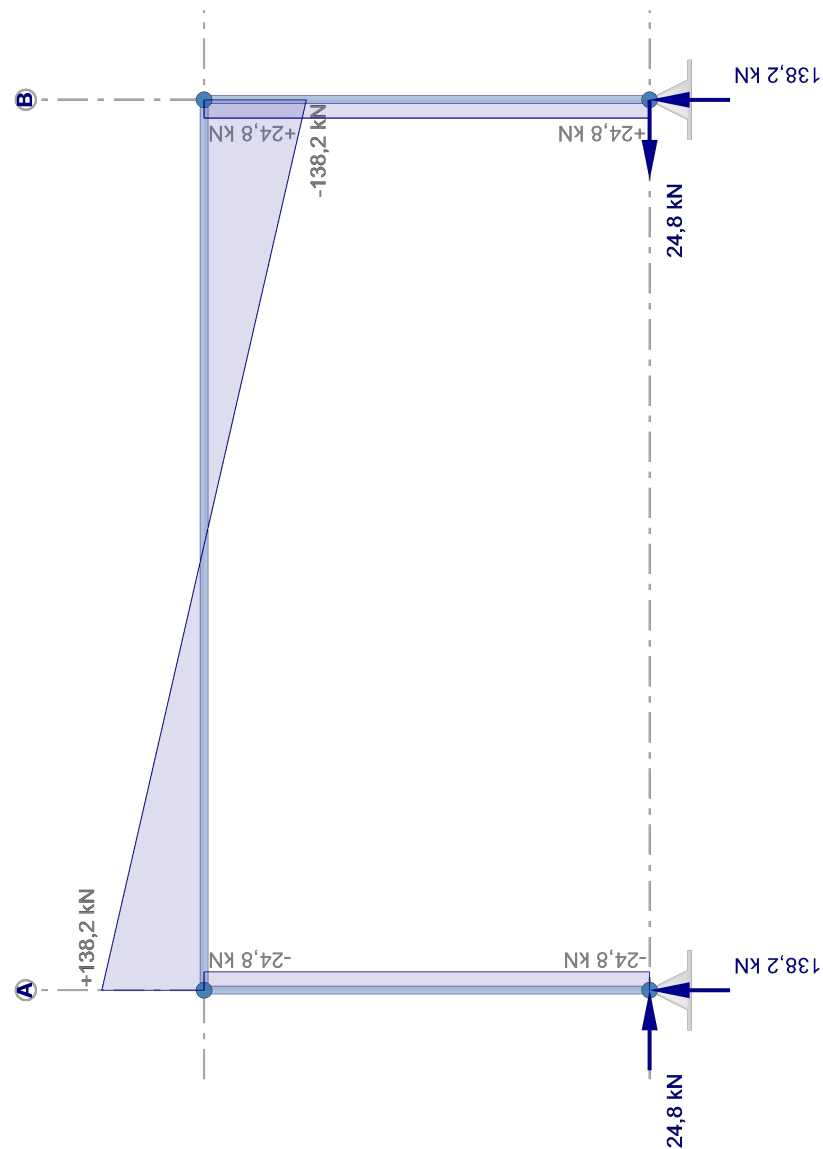
2.1 NODES - Global sway imperfection

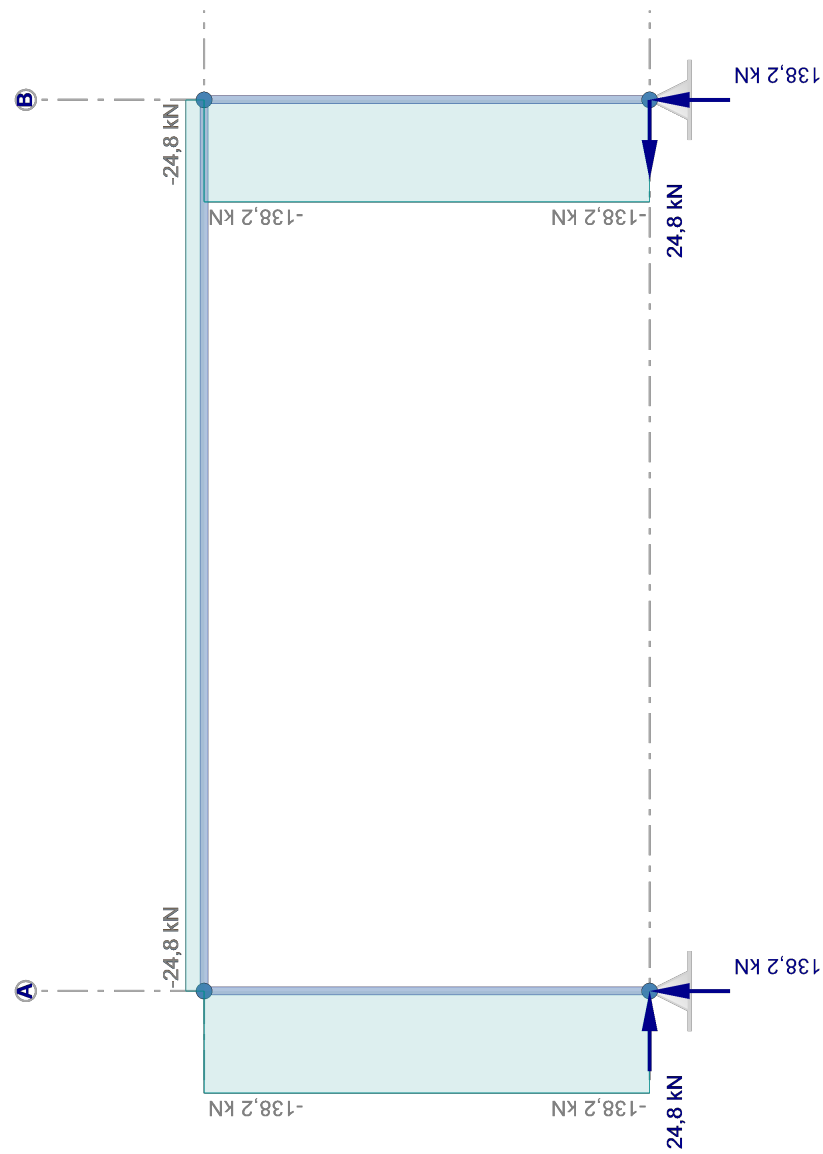
Node Number	1/200 in +X		1/200 in -X	
	X [mm]	Z [mm]	X [mm]	Z [mm]
1	0	0	0	0
2	5000	0	5000	0
3	13	2500	-13	2500
4	5013	2500	4988	2500

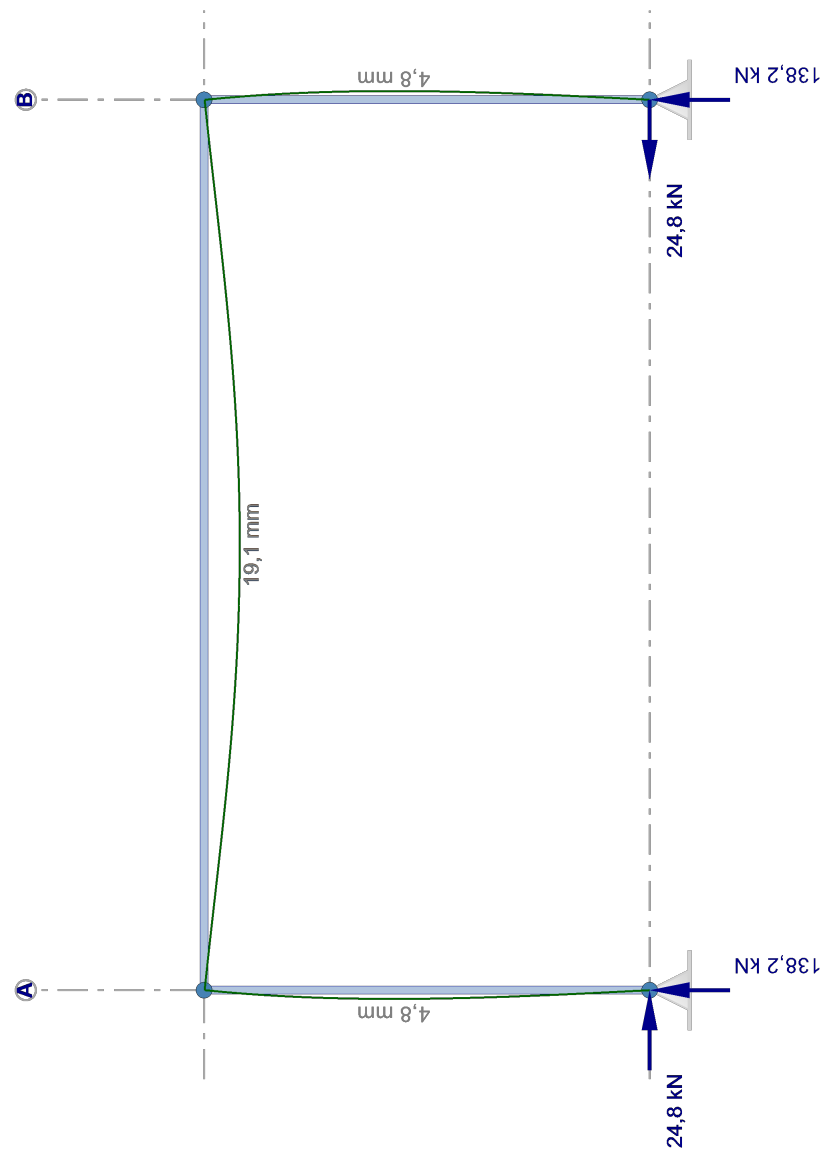
2.2 LOAD CASES

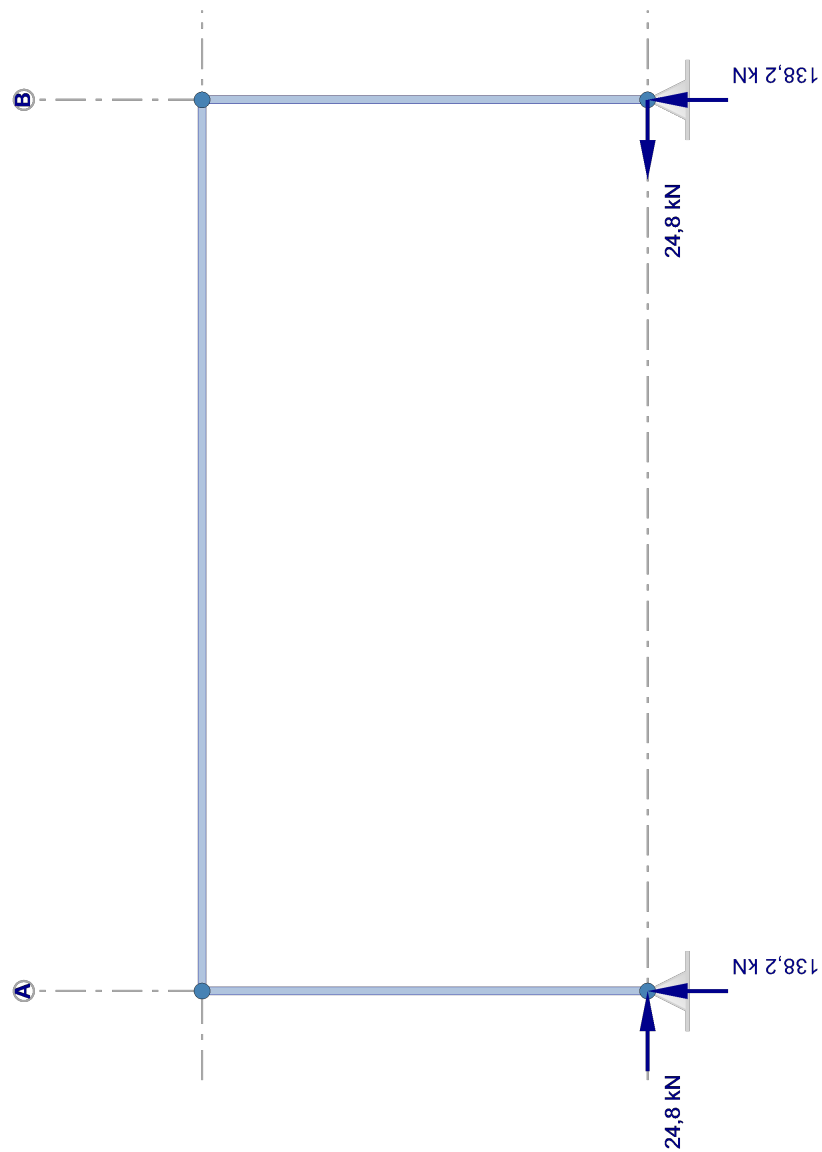
Geometric linear analysis

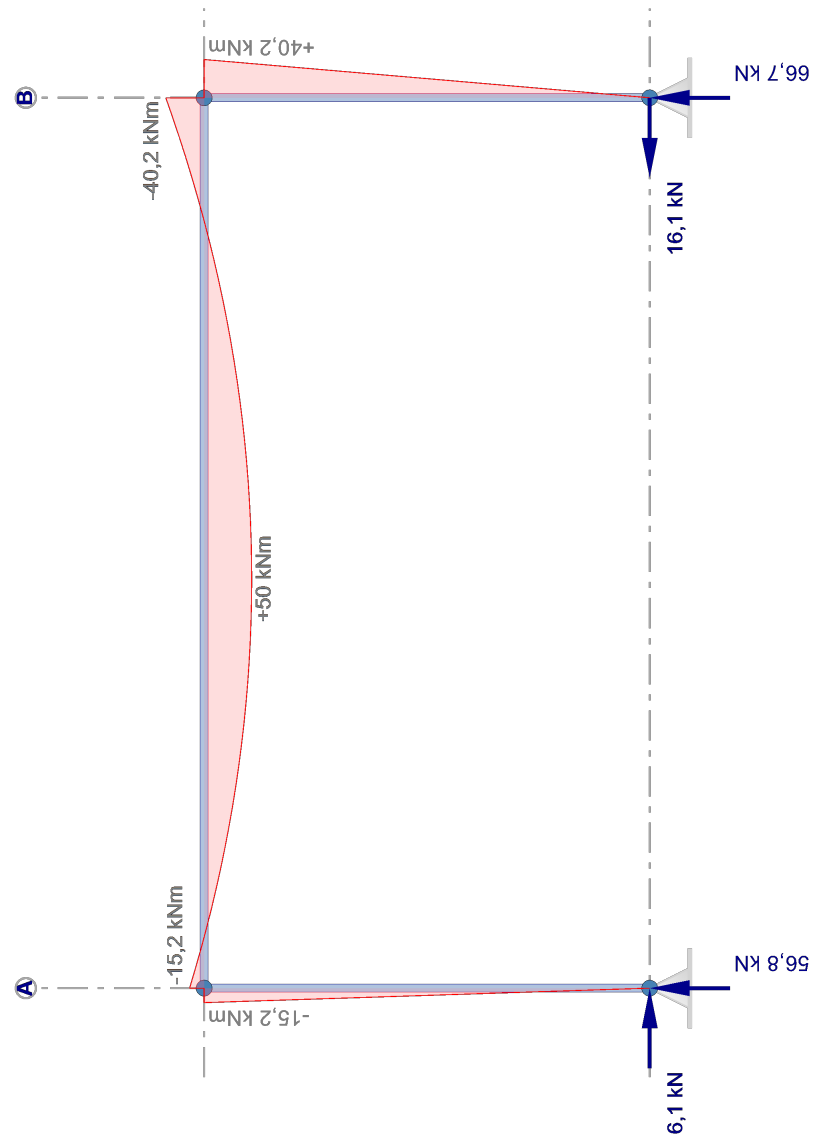


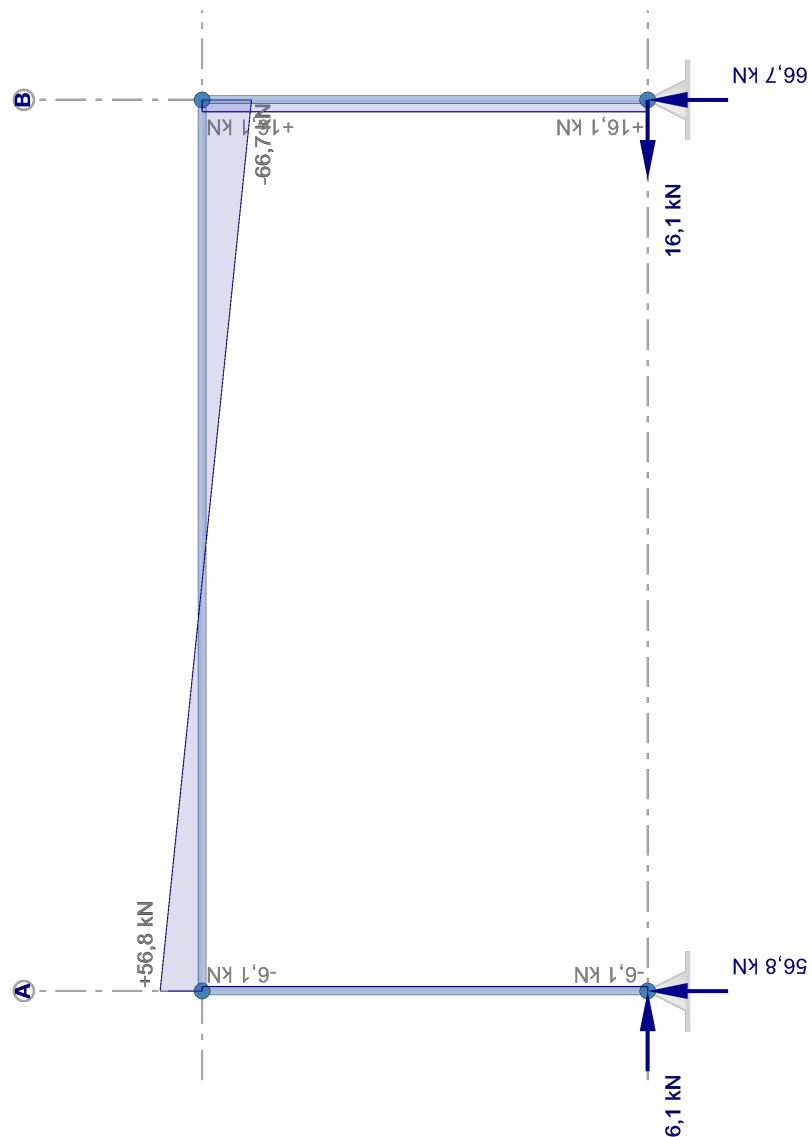


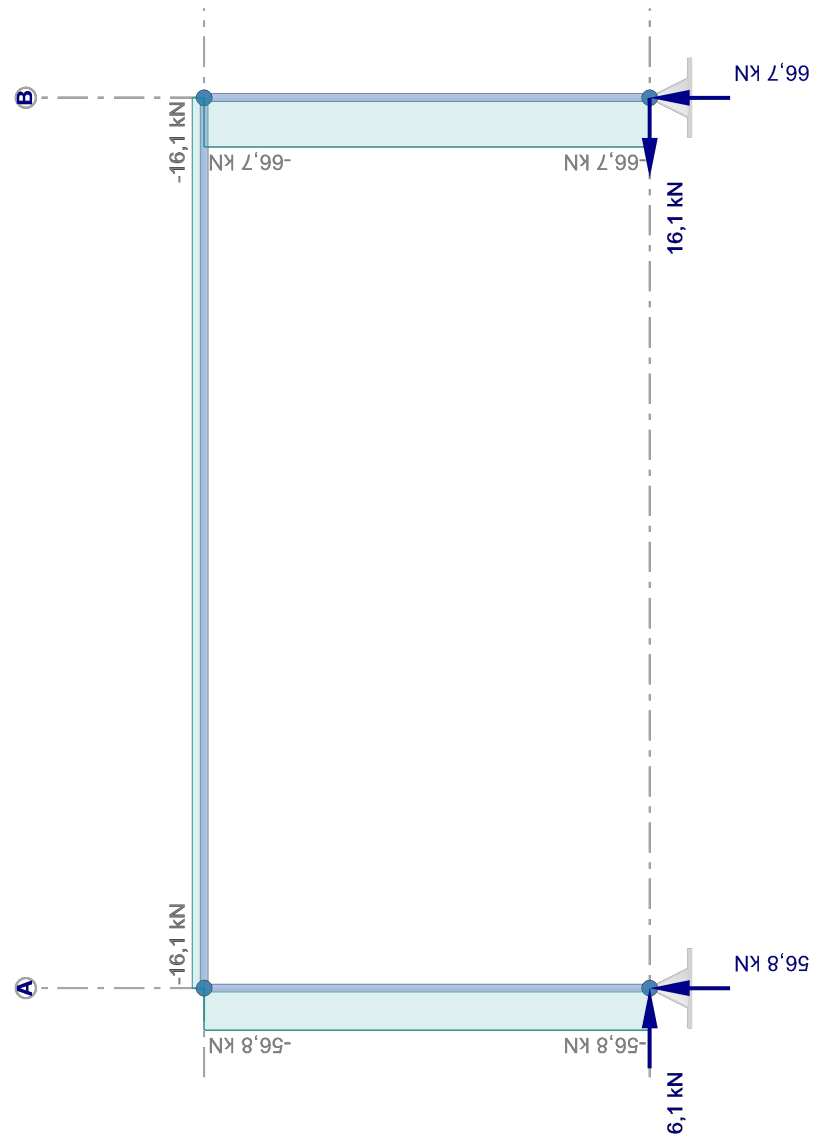
D-line - 1 Dead load**N-line - 1 Dead load**



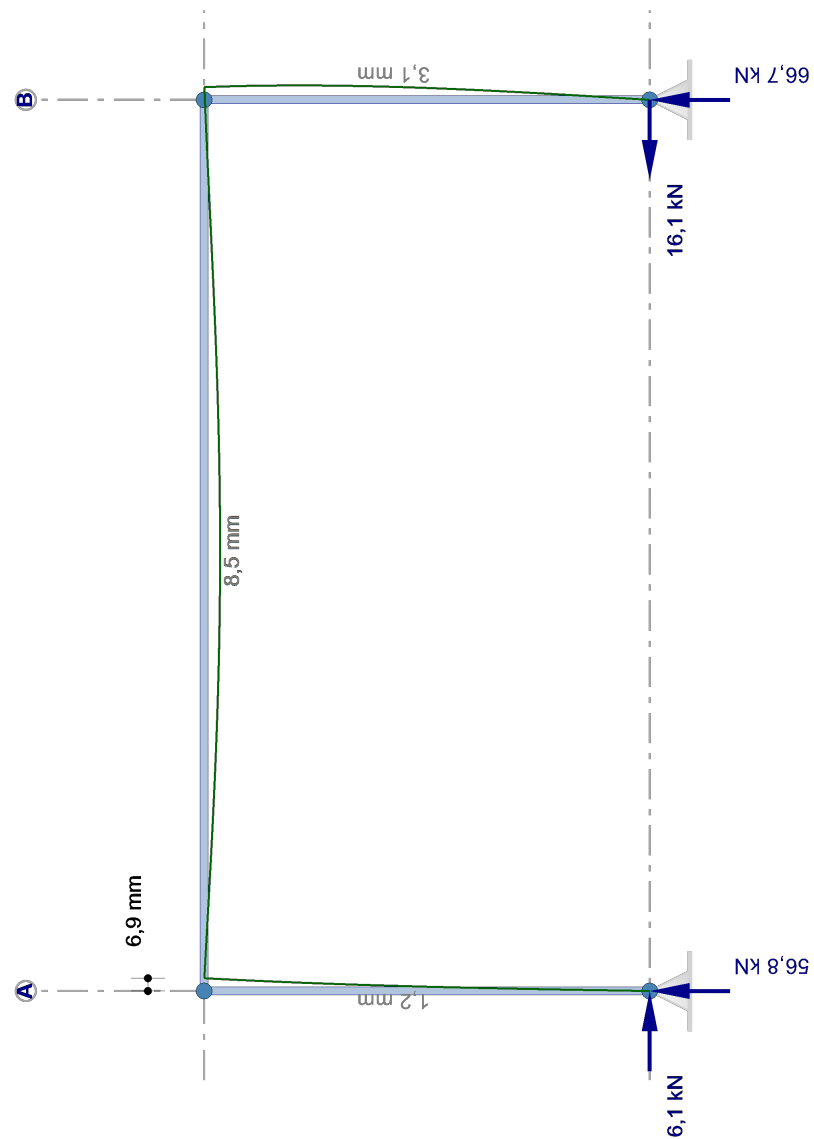
Displacements - 1 Dead load**Soil pressure - 1 Dead load**



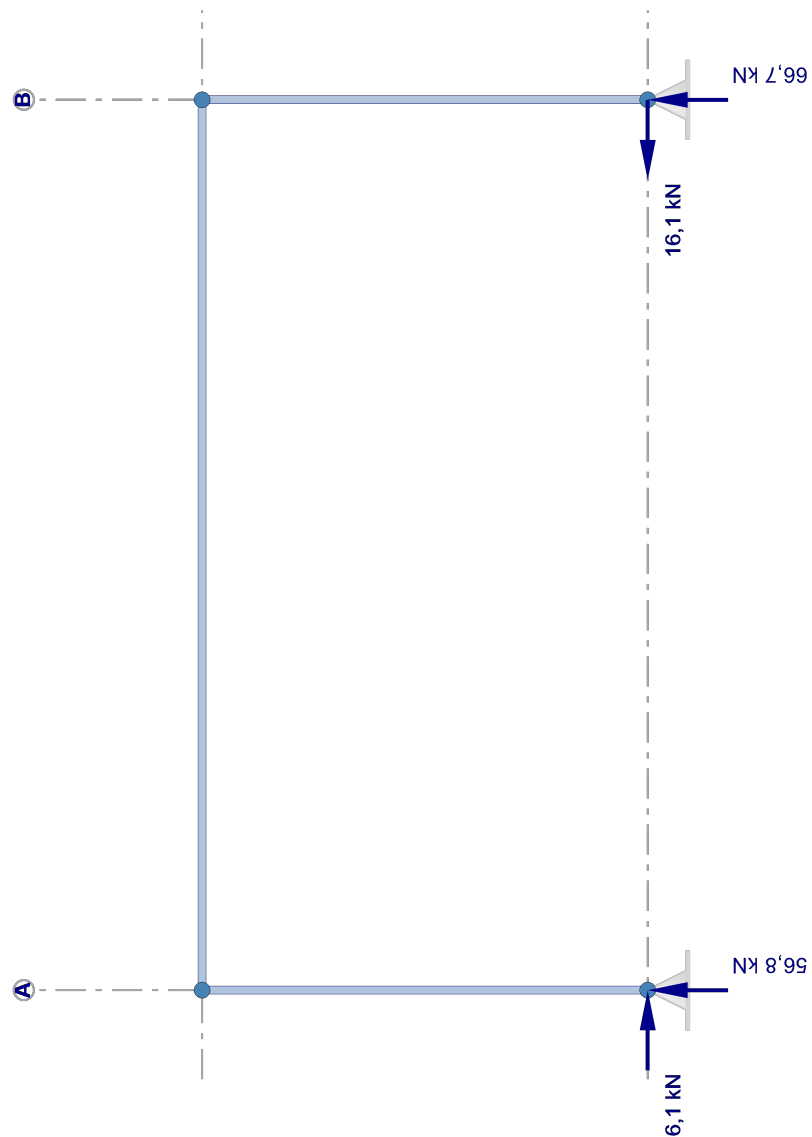
M-line - 2 Live load**D-line - 2 Live load**



N-line - 2 Live load



Displacements - 2 Live load



2.2.1 Node displacement

Node Number	Location
1	
2	
3	
4	
4	
3	
3	
1	
3	
4	

2.2.2 Envelope node

Node Number	Location
1	
2	
3	
4	
4	
3	
3	
1	
3	
4	

2.2.3 Reaction force

Node Number	Location
1	
2	
2	
1	

Soil pressure - 2 Live load**Displacements**

Node	dx [mm]	dz [mm]	dr [mrad]
1	0,0	0,0	4,9
2	0,0	0,0	-1,6
1	0,0	0,0	-4,9
2	0,0	0,0	-6,0
1	0,0	-0,3	-9,9
2	6,9	-0,1	-5,2
1	0,0	-0,3	9,9
2	6,9	-0,1	3,6
Minimum / maximum values			
1	0,0		
2	6,9		
1		-0,3	
2		0,0	
1			-9,9
1			9,9

Reaction displacements

Node	dx [mm]	dz [mm]	dr [mrad]
1	0,0	0,0	4,9
2	0,0	0,0	-1,6
1	0,0	0,0	-4,9
2	0,0	0,0	-6,0
1	0,0	-0,3	-9,9
2	6,9	-0,1	-5,2
1	0,0	-0,3	9,9
2	6,9	-0,1	3,6
Minimum / maximum values			
1	0,0		
2	6,9		
1		-0,3	
2		0,0	
1			-9,9
1			9,9

Reactions

Node	Fx [kN]	Fz [kN]	My [kNm]
1	24,827	138,214	
2	6,089	56,750	
1	-24,827	138,214	
2	-16,089	66,750	
Minimum / maximum values			
1	-24,827		
1	24,827		

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
1	2		56,750	
1	1		138,214	

2.2.4 Envelope reaction forces

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
1	1	24,827	138,214	
	2	6,089	56,750	
2	1	-24,827	138,214	
	2	-16,089	66,750	
Minimum / maximum values				
2	1	-24,827		
1	1	24,827		
1	2		56,750	
1	1		138,214	

2.2.5 Beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		138,214	-24,827	0,000
		3		-138,214	24,827	-62,067
	2	1		56,750	-6,089	0,000
		3		-56,750	6,089	-15,224
2	1	2		138,214	24,827	0,000
		4		-138,214	-24,827	62,067
	2	2		66,750	16,089	0,000
		4		-66,750	-16,089	40,224
3	1	3	2500	24,827	138,214	62,067
				-24,827	0,000	110,701
		4		-24,827	138,214	-62,067
	2	3		16,089	56,750	15,224
			2298	-16,089	0,000	49,970
		4		-16,089	66,750	-40,224

2.2.6 Envelope beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		138,214	-24,827	0,000
	2	1		56,750	-6,089	0,000
	1	3		-138,214	24,827	-62,067
	2	3		-56,750	6,089	-15,224
2	1	2		138,214	24,827	0,000
	2	2		66,750	16,089	0,000
	1	4		-138,214	-24,827	62,067
	2	4		-66,750	-16,089	40,224
3	1	3		24,827	138,214	62,067

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
3	2	3		16,089	56,750	15,224
	1		2500	-24,827	0,000	110,701
	1	4		-24,827	138,214	-62,067
	2	4		-16,089	66,750	-40,224

2.2.7 Beam stresses

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm ²]	sigma2 [N/mm ²]
1	1	1	0	138,214	-24,827	0,000	-25,5	-25,5
			2500	138,214	-24,827	-62,067	173,7	-224,6
1	2	1	0	56,750	-6,089	0,000	-10,5	-10,5
			2500	56,750	-6,089	-15,224	38,4	-59,3
2	1	2	0	138,214	24,827	0,000	-25,5	-25,5
			2500	138,214	24,827	62,067	-224,6	173,7
2	2	2	0	66,750	16,089	0,000	-12,3	-12,3
			2500	66,750	16,089	40,224	-141,4	116,8
3	1	3	0	24,827	138,214	-62,067	94,9	-99,2
			2500	24,827	0,000	110,701	-175,3	171,0
			5000	24,827	-138,214	-62,067	94,9	-99,2
3	2	3	0	16,089	56,750	-15,224	22,4	-25,2
			2298	16,089	0,000	49,970	-79,6	76,8
			5000	16,089	-66,750	-40,224	61,5	-64,3

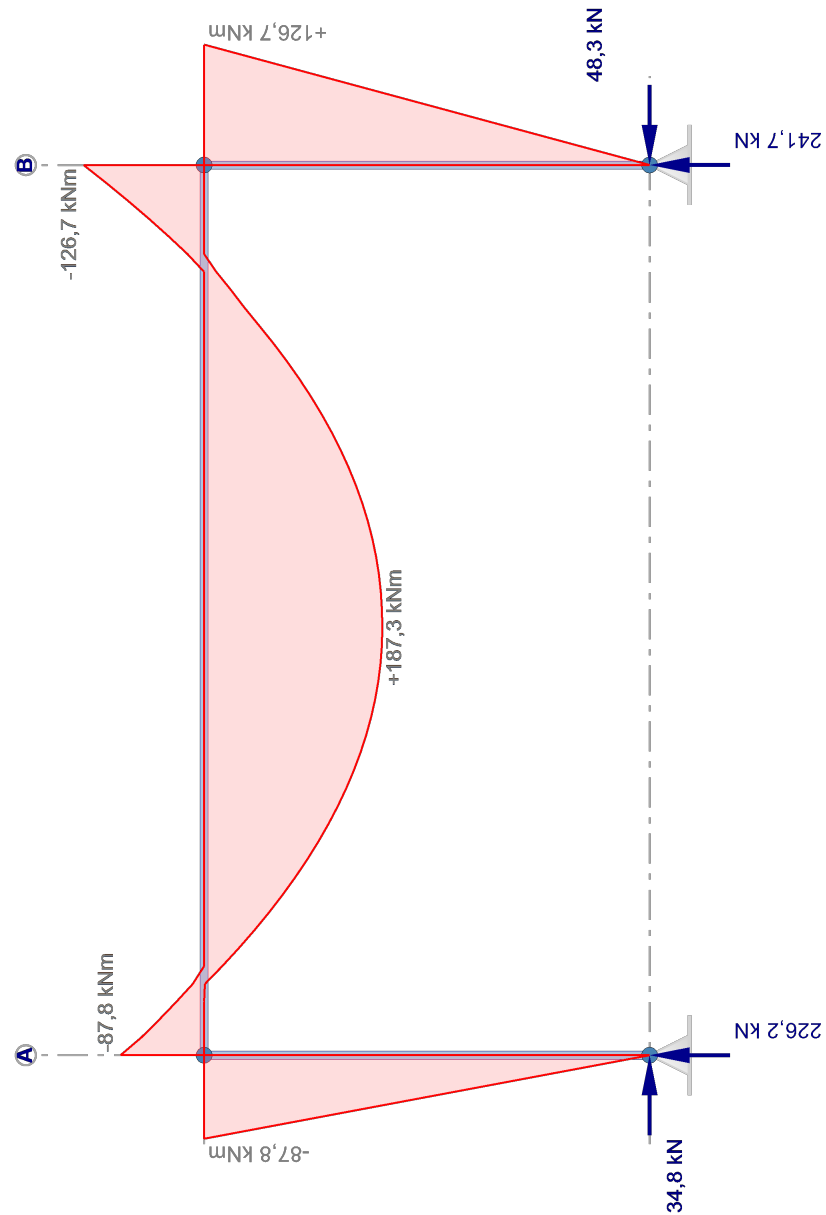
2.3 ULTIMATE LIMIT STATES (ULS)

2.3.1 Load combinations

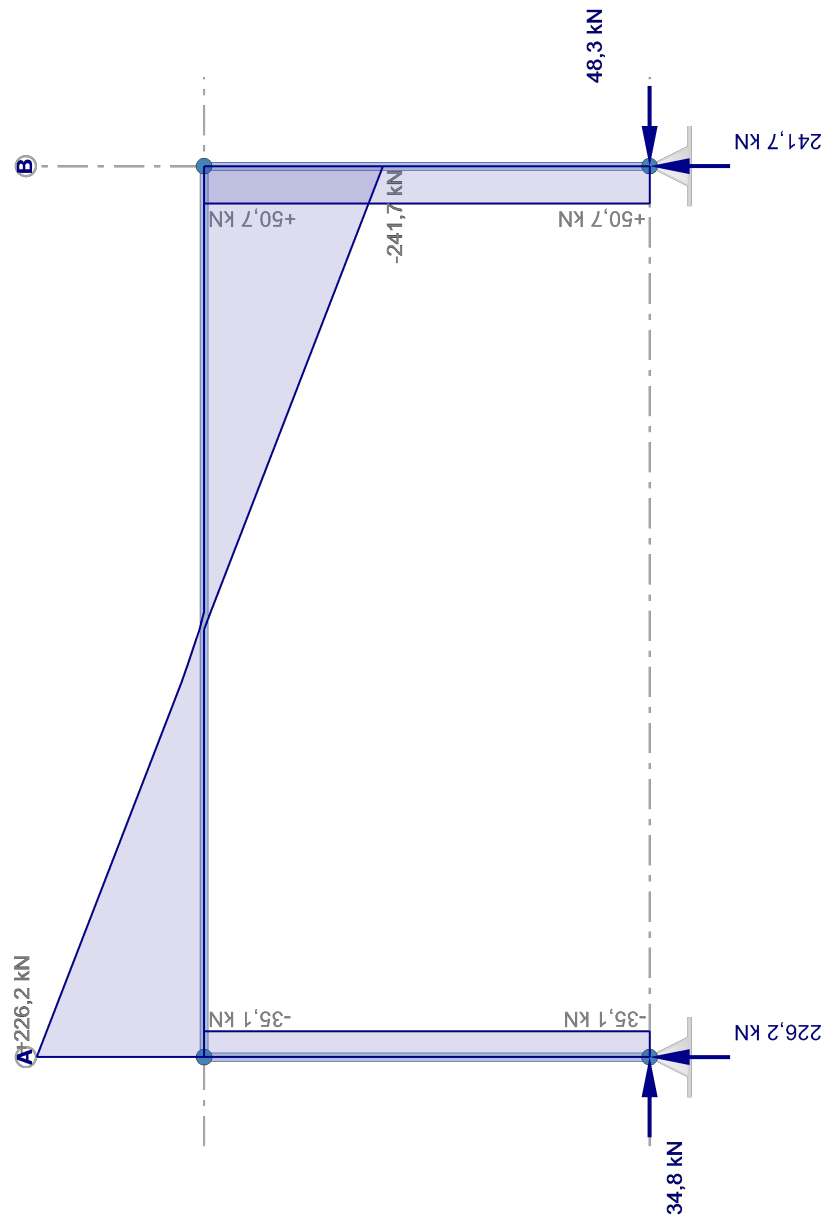
Geometric nonlinear analysis

Combination Number	Description	Type
1	Combination1 (6.10a)	ULS
1	Combination1 (6.10a)	ULS
2	Combination2 (6.10b)	ULS
2	Combination2 (6.10b)	ULS

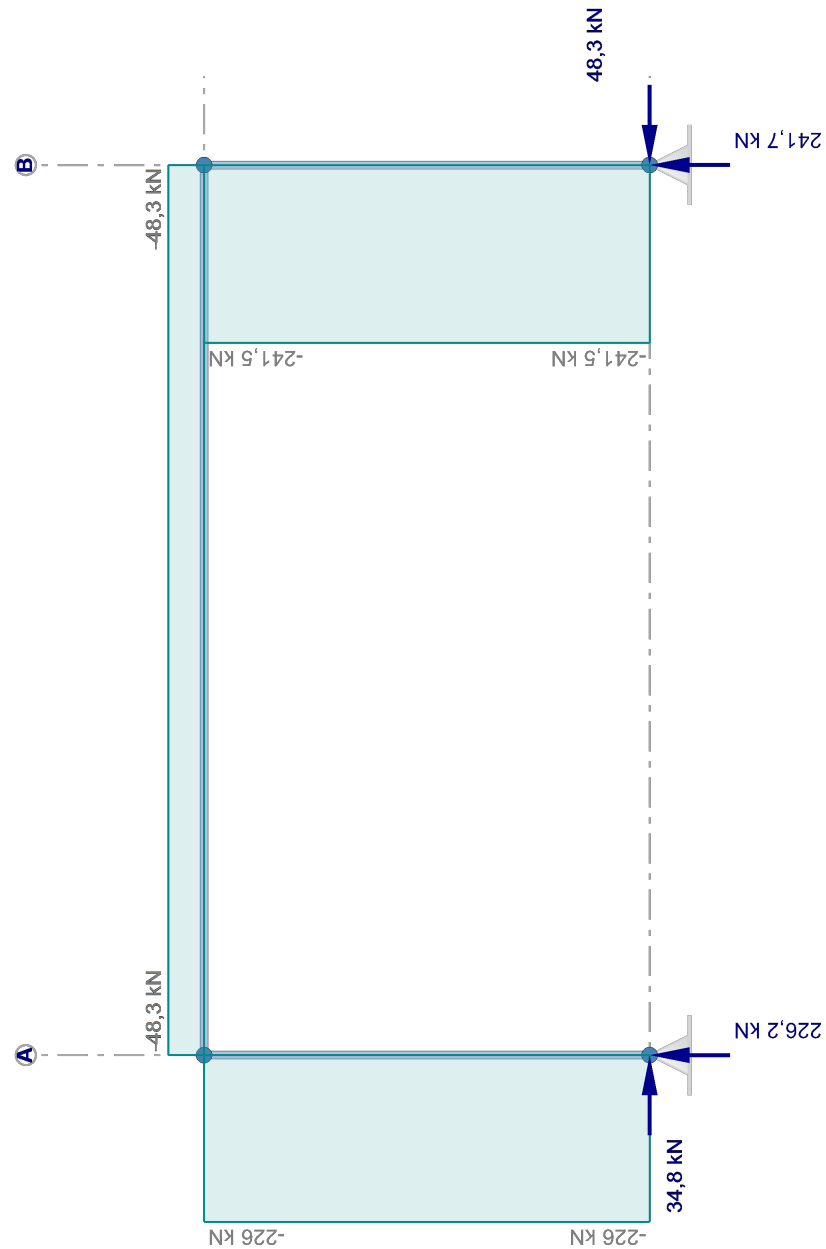
Combination Number	Load case (ψ x γ)				
	1	2			
1	1,00x1,22	0,40x1,35			
1	1,00x1,22	0,40x1,35			
2	1,00x1,08	1,00x1,35			
2	1,00x1,08	1,00x1,35			



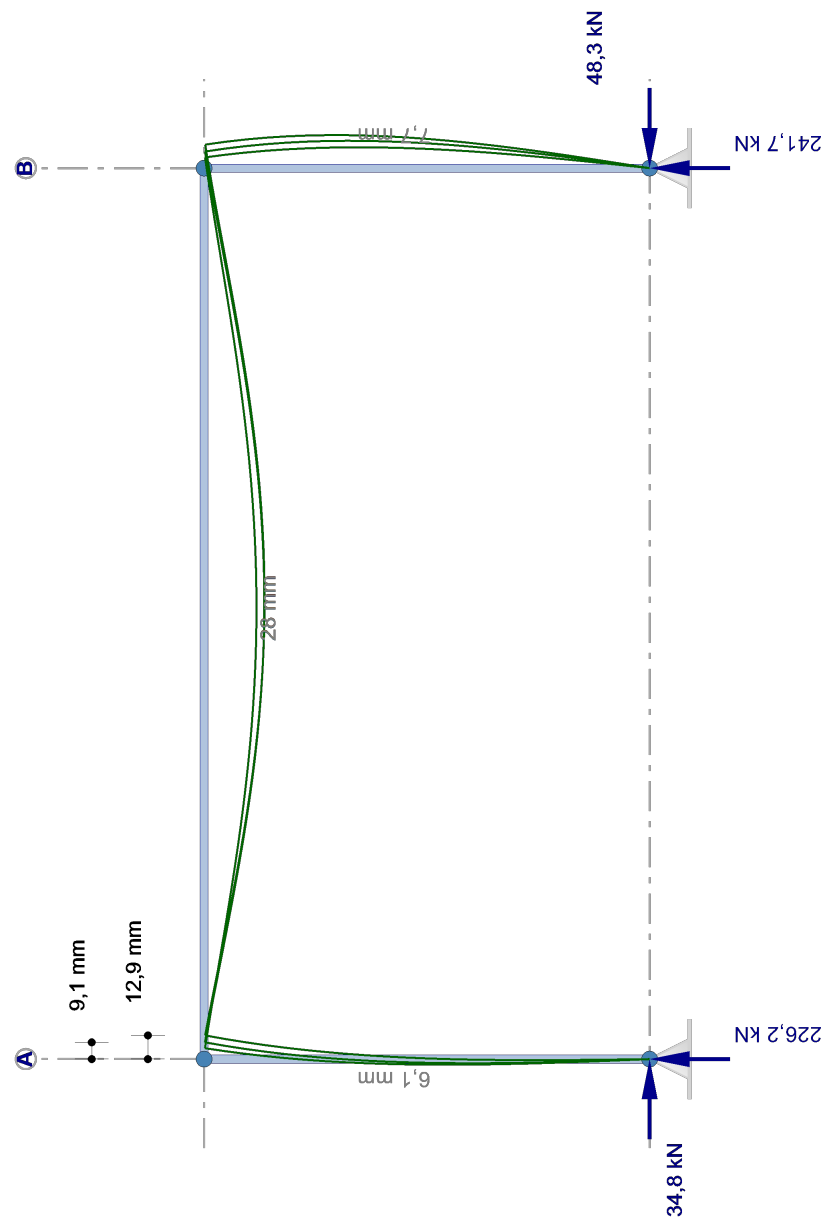
Envelop M-line



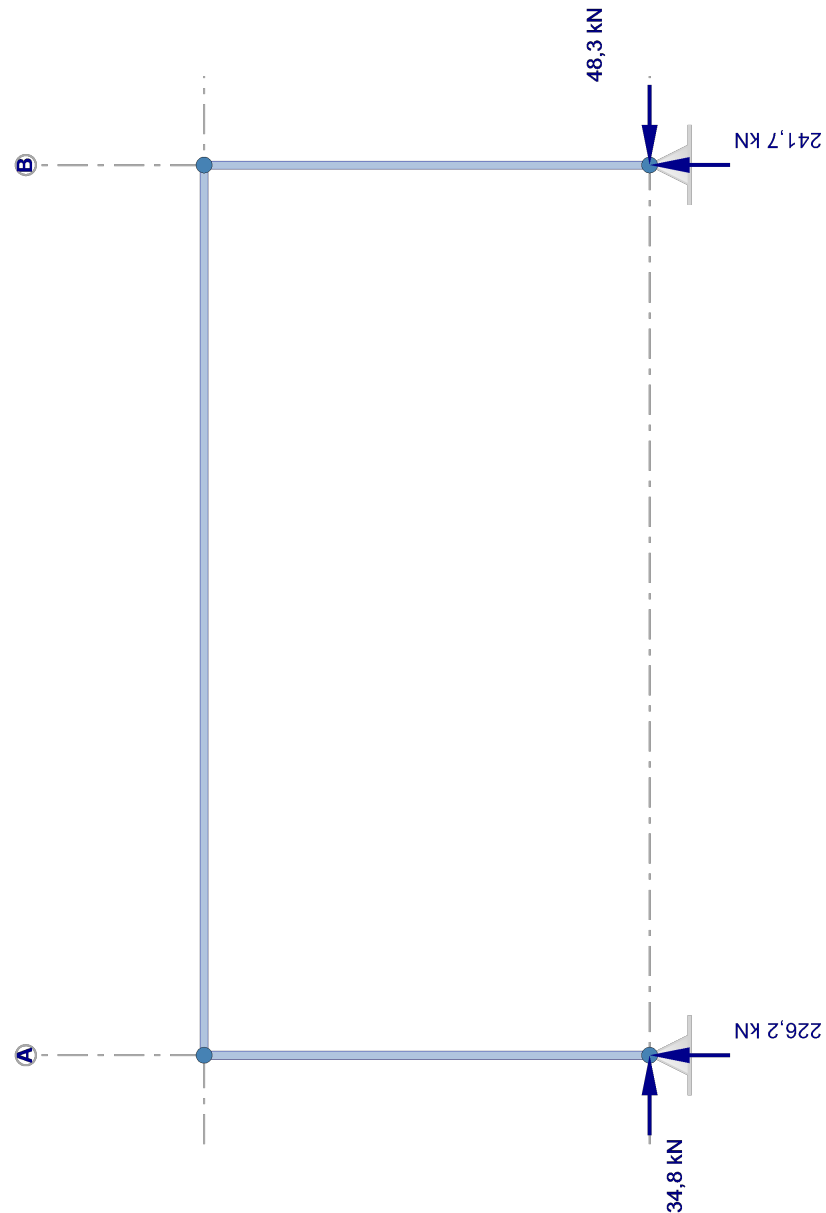
Envelop D-line



Envelop N-line



Envelop displacements



2.3.2 Node displace

Node Number	Comb Nu
1	
2	
3	
4	
1	
3	
4	
1	
3	
4	

2.3.3 Envelope node

Node Number	Comb Nu
1	
2	
3	
4	
1	
3	
4	
1	
3	

Envelop soil pressure

Elements

Combination number	dx [mm]	dz [mm]	dr [mrad]
1	0,0	0,0	4,2
1	0,0	0,0	5,9
2	0,0	0,0	1,5
2	0,0	0,0	3,6
1	0,0	0,0	-10,6
1	0,0	0,0	-8,8
2	0,0	0,0	-15,5
2	0,0	0,0	-13,5
1	5,9	-0,5	-15,3
1	2,7	-0,4	-14,9
2	12,9	-0,6	-18,3
2	9,1	-0,5	-17,9
1	5,8	-0,5	14,0
1	2,6	-0,4	14,3
2	12,8	-0,6	15,5
2	9,0	-0,5	15,9
Minimum / maximum values			
2	0,0		
2	12,9		
2		-0,6	
1		0,0	
2			-18,3
2			15,9

The displacements

Combination number	dx [mm]	dz [mm]	dr [mrad]
1	0,0	0,0	4,2
1	0,0	0,0	5,9
2	0,0	0,0	1,5
2	0,0	0,0	3,6
1	0,0	0,0	-10,6
1	0,0	0,0	-8,8
2	0,0	0,0	-15,5
2	0,0	0,0	-13,5
1	2,7	-0,4	-14,9
2	12,9	-0,6	-18,3
1	5,8	-0,5	14,0
1	2,6	-0,4	14,3
2	12,8	-0,6	15,5
2	9,0	-0,5	15,9
Minimum / maximum values			
2	0,0		
2	12,9		
2		-0,6	
1		0,0	
2			-18,3

Node Number	Combination Number	dx [mm]	dz [mm]	dr [mrad]
4	2			15,9

2.3.4 Reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
1	1	33,380	197,782	
	1	33,417	200,060	
	2	34,728	223,528	
	2	34,829	226,204	
2	1	-38,780	206,151	
	1	-38,817	203,873	
	2	-48,228	241,740	
	2	-48,329	239,063	
Minimum / maximum values				
2	2	-48,329		
1	2	34,829		
1	1		197,782	
2	2		241,740	

2.3.5 Envelope reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
1	1	33,380	197,782	
	2	34,829	226,204	
2	1	-38,780	206,151	
	1	-38,817	203,873	
	2	-48,228	241,740	
	2	-48,329	239,063	
Minimum / maximum values				
2	2	-48,329		
1	2	34,829		
1	1		197,782	
2	2		241,740	

2.3.6 Beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		197,946	-31,922	0,000
		3		-197,946	31,922	-79,804
		1		199,890	-34,199	0,000
		3		-199,890	34,199	-85,498
	2	1		223,699	-32,458	0,000
		3		-223,699	32,458	-81,147
		1		226,027	-35,136	0,000
		3		-226,027	35,136	-87,841
2	1	2		205,954	40,291	0,000

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
2	1	4		-205,954	-40,291	100,726
		2		204,065	38,012	0,000
		4		-204,065	-38,012	95,031
		2		241,496	50,669	0,000
	2	4		-241,496	-50,669	126,675
		2		239,302	47,995	0,000
		4		-239,302	-47,995	119,987
		4				
3	1	3		38,780	197,782	79,804
			2448	-38,780	0,000	162,301
		4		-38,780	206,151	-100,726
		3		38,817	200,060	85,498
	1		2476	-38,817	0,000	162,216
		4		-38,817	203,873	-95,031
		3		48,228	223,528	81,147
			2402	-48,228	0,000	187,327
	2	4		-48,228	241,739	-126,675
		3		48,329	226,205	87,841
			2431	-48,329	0,000	187,100
		4		-48,329	239,063	-119,987
	2					

2.3.7 Envelope beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		197,946	-31,922	0,000
	2	1		226,027	-35,136	0,000
	1	3		-197,946	31,922	-79,804
	2	3		-226,027	35,136	-87,841
2	1	2		204,065	38,012	0,000
	2	2		241,496	50,669	0,000
	1	4		-204,065	-38,012	95,031
	2	4		-241,496	-50,669	126,675
3	1	3		38,780	197,782	79,804
	2	3		48,329	226,205	87,841
	2		2402	-48,228	0,000	187,327
	1	4		-38,780	206,151	-100,726
	1	4		-38,817	203,873	-95,031
	2	4		-48,228	241,739	-126,675
	2	4		-48,329	239,063	-119,987

2.3.8 Beam stresses

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm²]	sigma2 [N/mm²]
1	1	1	0	197,946	-31,922	0,000	-36,5	-36,5
			2500	197,946	-31,922	-79,806	219,6	-292,6
1	1	1	0	199,890	-34,199	0,000	-36,8	-36,8
			2500	199,890	-34,199	-85,497	237,5	-311,2
1	2	1	0	223,699	-32,458	0,000	-41,2	-41,2
			2500	223,699	-32,458	-81,146	219,2	-301,6

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm ²]	sigma2 [N/mm ²]
1	2	1	0	226,027	-35,136	0,000	-41,6	-41,6
			2500	226,027	-35,136	-87,840	240,2	-323,5
2	1	2	0	205,954	40,291	0,000	-37,9	-37,9
			2500	205,954	40,291	100,727	-361,2	285,3
2	1	2	0	204,065	38,012	0,000	-37,6	-37,6
			2500	204,065	38,012	95,031	-342,6	267,4
2	2	2	0	241,496	50,669	0,000	-44,5	-44,5
			2500	241,496	50,669	126,672	-451,0	362,0
2	2	2	0	239,302	47,995	0,000	-44,1	-44,1
			2500	239,302	47,995	119,987	-429,1	341,0
3	1	3	0	38,780	197,782	-79,804	121,5	-128,2
			2448	38,780	0,000	162,301	-257,2	250,5
			5000	38,780	-206,151	-100,726	154,2	-160,9
3	1	3	0	38,817	200,060	-85,498	130,4	-137,1
			2476	38,817	0,000	162,216	-257,1	250,4
			5000	38,817	-203,873	-95,031	145,3	-152,0
3	2	3	0	48,228	223,528	-81,147	122,7	-131,1
			2402	48,228	0,000	187,327	-297,2	288,8
			5000	48,228	-241,739	-126,675	194,0	-202,3
3	2	3	0	48,329	226,205	-87,841	133,2	-141,6
			2431	48,329	0,000	187,100	-296,9	288,5
			5000	48,329	-239,063	-119,987	183,5	-191,9

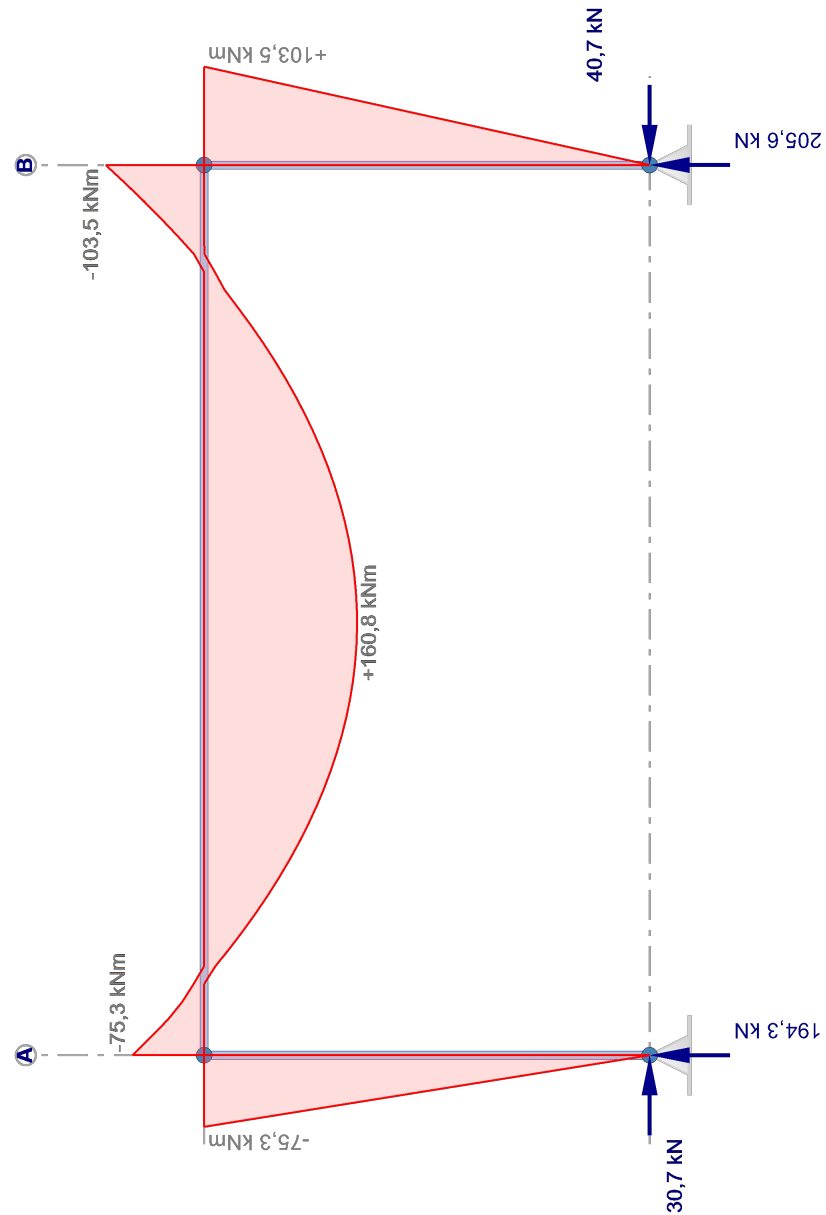
2.4 SERVICE LIMIT STATES (SLS)

2.4.1 Load combinations

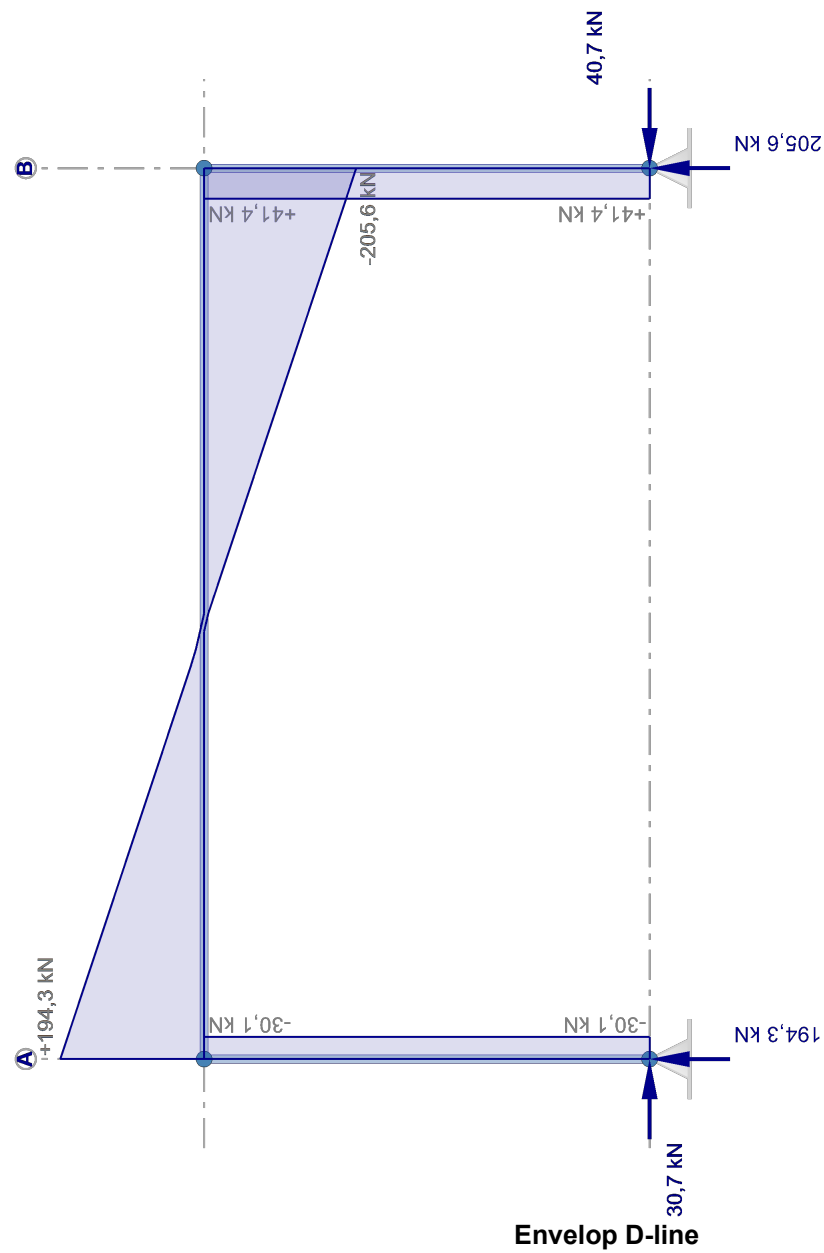
Geometric nonlinear analysis

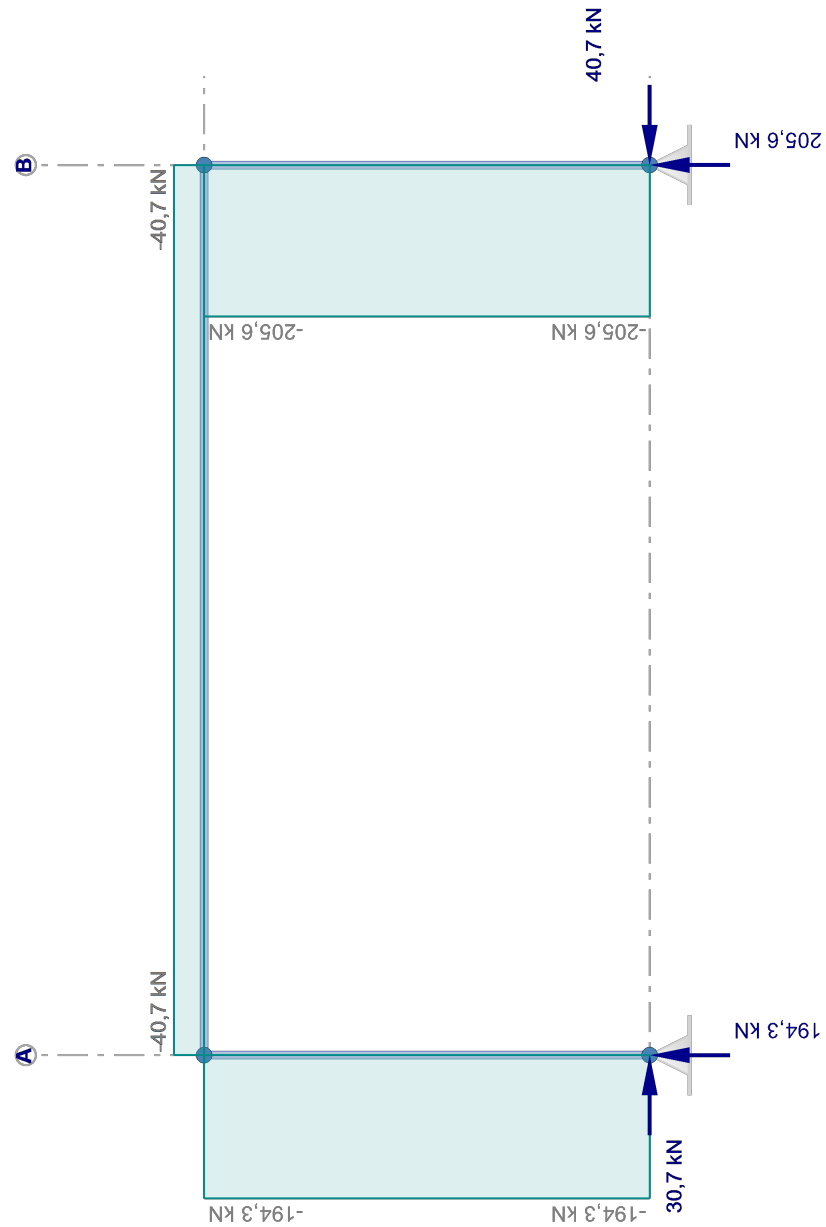
Combination Number	Description	Type
3	SLS Permanent	SLS Permanent
4	SLS Quasi permanent	SLS Quasi permanent
5	Combination	SLS

Combination Number	Load case ($\psi \times \gamma$)				
	1	2			
3	1,00x1,00				
4	1,00x1,00	0,30x1,00			
5	1,00x1,00	1,00x1,00			

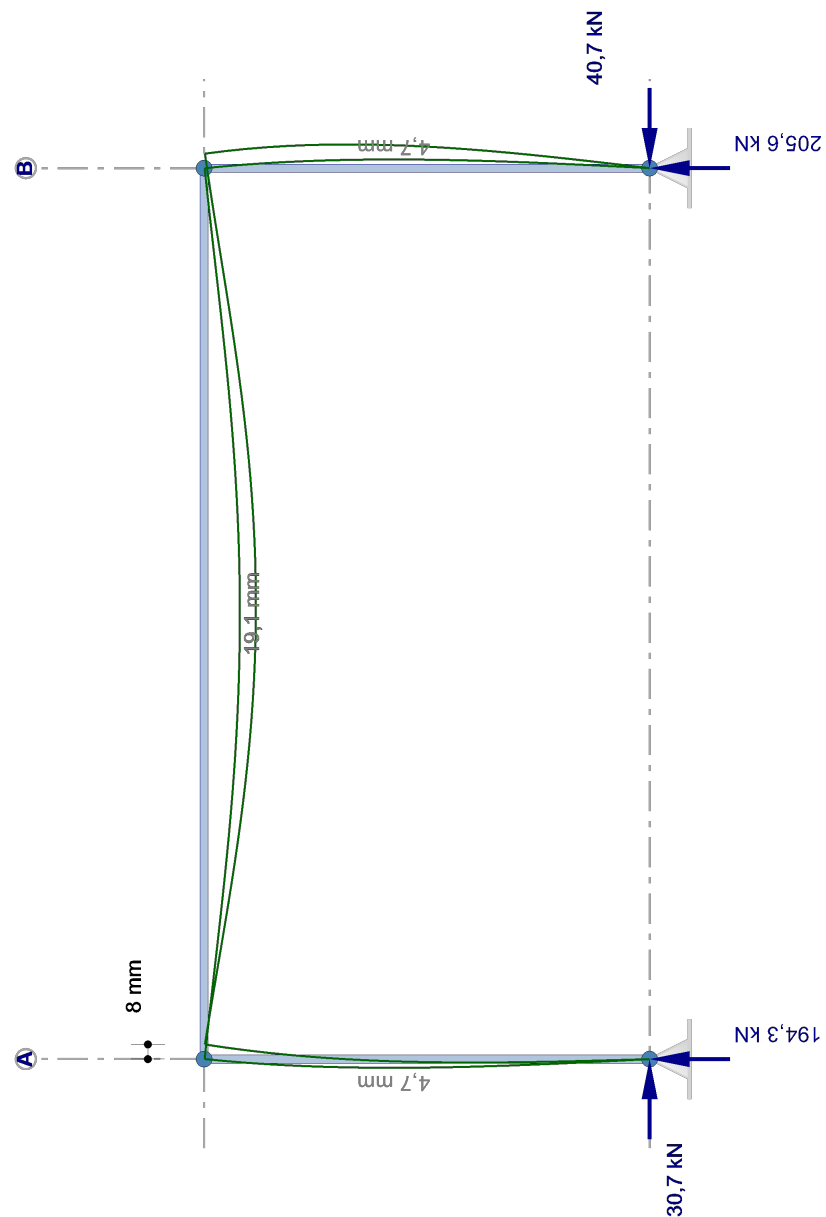


Envelop M-line

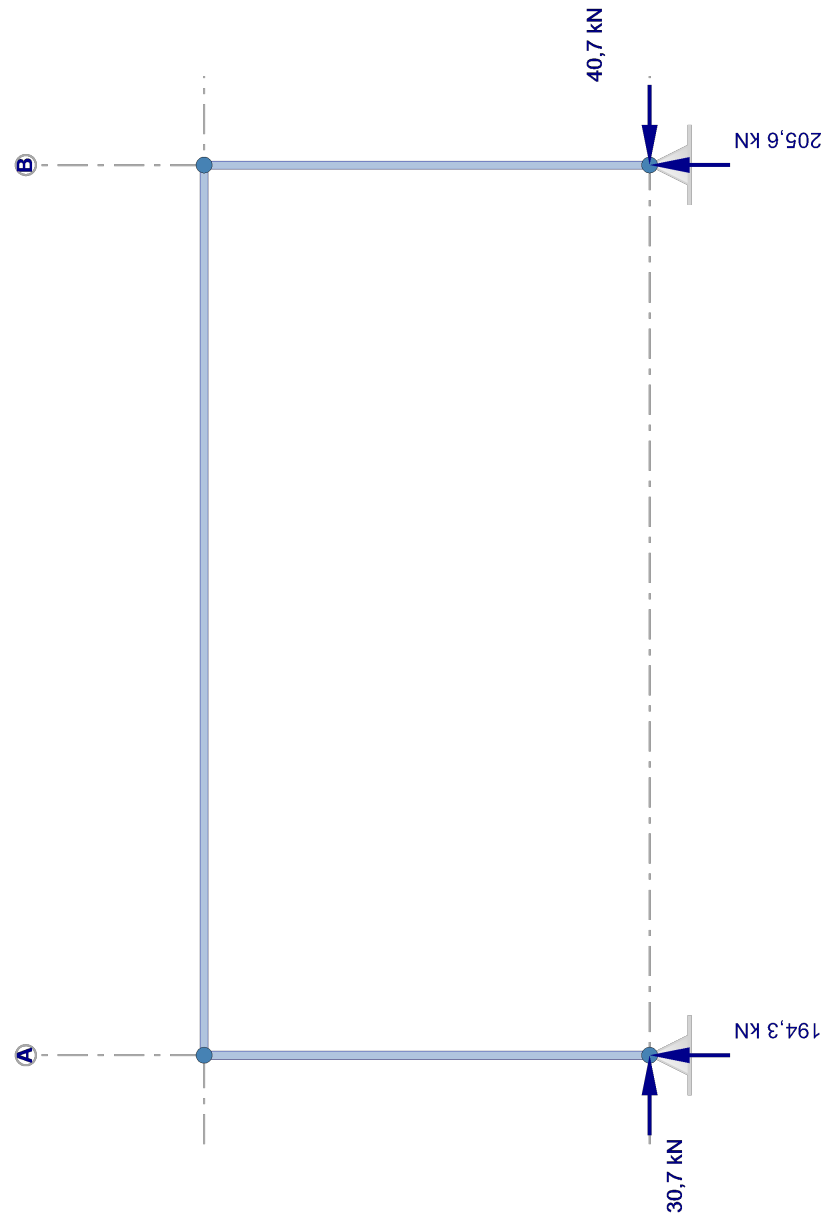




Envelop N-line



Envelop displacements



2.4.2 Node displace

Node Number	Comb Nu
1	
2	
3	
4	
4	
3	
4	
1	
3	
4	

2.4.3 Envelope node

Node Number	Comb Nu
1	
2	
3	
4	
4	
3	
4	
1	
3	
4	

2.4.4 Reaction force

Node Number	Comb Nu
1	

Envelop soil pressure**ements**

ination umber	dx [mm]	dz [mm]	dr [mrad]
3	0,0	0,0	5,0
4	0,0	0,0	4,4
5	0,0	0,0	3,0
3	0,0	0,0	-5,0
4	0,0	0,0	-6,9
5	0,0	0,0	-11,6
3	0,0	-0,3	-10,0
4	2,3	-0,3	-11,6
5	8,0	-0,4	-15,4
3	0,0	-0,3	10,0
4	2,3	-0,3	11,1
5	7,9	-0,5	13,6
Minimum / maximum values			
3	0,0		
5	8,0		
5		-0,5	
3		0,0	
5			-15,4
5			13,6

le displacements

ination umber	dx [mm]	dz [mm]	dr [mrad]
3	0,0	0,0	5,0
5	0,0	0,0	3,0
3	0,0	0,0	-5,0
5	0,0	0,0	-11,6
3	0,0	-0,3	-10,0
5	8,0	-0,4	-15,4
3	0,0	-0,3	10,0
5	7,9	-0,5	13,6
Minimum / maximum values			
3	0,0		
5	8,0		
5		-0,5	
3		0,0	
5			-15,4
5			13,6

es

ination umber	Fx [kN]	Fz [kN]	My [kNm]
3	24,745	138,214	
4	26,550	155,095	
5	30,739	194,331	

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
2	3	-24,745	138,214	
	4	-29,550	158,384	
	5	-40,739	205,597	
Minimum / maximum values				
2	5	-40,739		
1	5	30,739		
1	3		138,214	
2	5		205,597	

2.4.5 Envelope reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
1	3	24,745	138,214	
	5	30,739	194,331	
2	3	-24,745	138,214	
	5	-40,739	205,597	
Minimum / maximum values				
2	5	-40,739		
1	5	30,739		
1	3		138,214	
2	5		205,597	

2.4.6 Beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	3	1		138,214	-24,744	0,000
		3		-138,214	24,744	-61,860
	4	1		155,095	-26,406	0,000
		3		-155,095	26,406	-66,014
	5	1		194,331	-30,121	0,000
		3		-194,331	30,121	-75,303
2	3	2		138,214	24,744	0,000
		4		-138,214	-24,744	61,860
	4	2		158,384	29,694	0,000
		4		-158,384	-29,694	74,236
	5	2		205,597	41,387	0,000
		4		-205,597	-41,387	103,466
3	3	3	2500	24,745	138,214	61,860
				-24,745	0,000	110,908
		4	2474	-24,745	138,214	-61,860
				29,550	155,095	66,014
	4	3	2474	-29,550	0,000	125,821
				-29,550	158,384	-74,236
		4	2430	40,739	194,332	75,303
				-40,739	0,000	160,769
	5	3	2430	-40,739	205,597	-103,466
		4				

2.4.7 Envelope beam forces

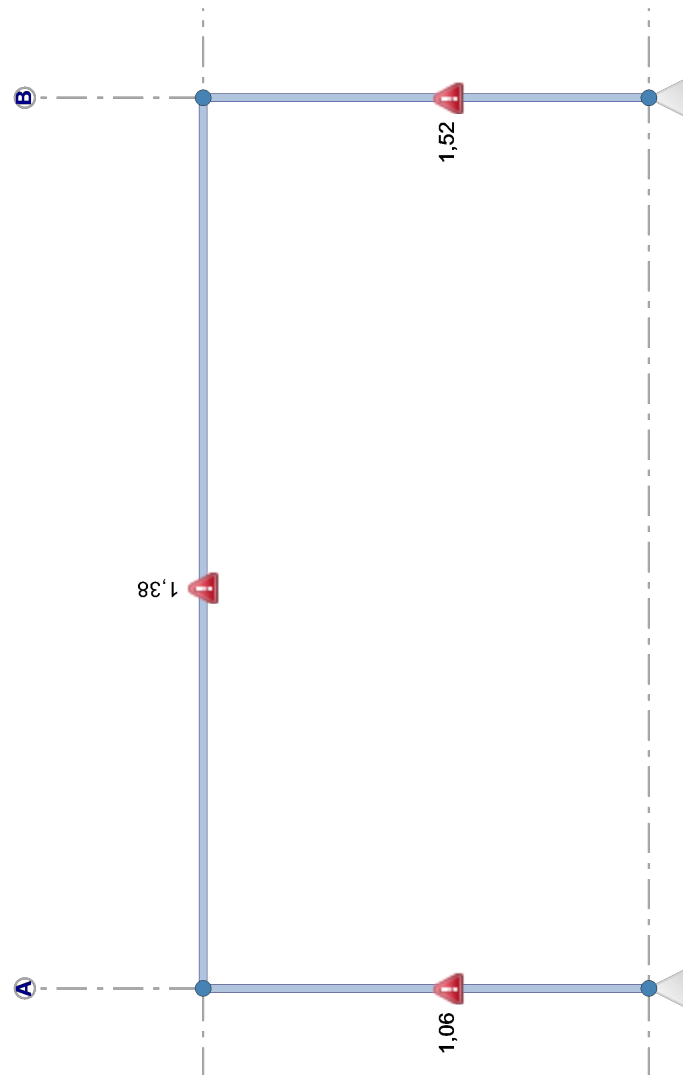
Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	3	1		138,214	-24,744	0,000
	5	1		194,331	-30,121	0,000
	3	3		-138,214	24,744	-61,860
	5	3		-194,331	30,121	-75,303
2	3	2		138,214	24,744	0,000
	5	2		205,597	41,387	0,000
	3	4		-138,214	-24,744	61,860
	5	4		-205,597	-41,387	103,466
3	3	3		24,745	138,214	61,860
	5	3		40,739	194,332	75,303
	5		2430	-40,739	0,000	160,769
	3	4		-24,745	138,214	-61,860
	5	4		-40,739	205,597	-103,466

2.4.8 Beam stresses

Beam Number	Comb. Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm ²]	sigma2 [N/mm ²]
1	3	1	0	138,214	-24,744	0,000	-25,5	-25,5
			2500	138,214	-24,744	-61,860	173,1	-224,0
1	4	1	0	155,095	-26,406	0,000	-28,6	-28,6
			2500	155,095	-26,406	-66,014	183,3	-240,4
1	5	1	0	194,331	-30,121	0,000	-35,8	-35,8
			2500	194,331	-30,121	-75,303	205,9	-277,5
2	3	2	0	138,214	24,744	0,000	-25,5	-25,5
			2500	138,214	24,744	61,860	-224,0	173,1
2	4	2	0	158,384	29,694	0,000	-29,2	-29,2
			2500	158,384	29,694	74,236	-267,4	209,1
2	5	2	0	205,597	41,387	0,000	-37,9	-37,9
			2500	205,597	41,387	103,466	-369,9	294,2
3	3	3	0	24,745	138,214	-61,860	94,6	-98,9
			2500	24,745	0,000	110,908	-175,6	171,3
			5000	24,745	-138,214	-61,860	94,6	-98,9
3	4	3	0	29,550	155,095	-66,014	100,7	-105,8
			2474	29,550	0,000	125,821	-199,4	194,2
			5000	29,550	-158,384	-74,236	113,6	-118,7
3	5	3	0	40,739	194,332	-75,303	114,2	-121,3
			2430	40,739	0,000	160,769	-255,0	247,9
			5000	40,739	-205,597	-103,466	158,3	-165,4

2.5 EN1993 CHECKS

The check of the steel members in the ultimate limit state according to EN 1993-1-1 is based on a geometrical non-linear analysis (second order analysis) using the imperfections as given in 5.3.2. (a) global initial sway imperfections, see Figure 5.2)



Beam Number	Profile	Combination Number	Class	Article	U.C.
1	HEB160	2	1	6.2.4	0,18
		2	1	6.2.5	1,06
		2	1	6.2.6	0,15
		2	1	6.2.8	1,06
		2	1	6.2.9.1	1,06
		1	1	6.3.2.1	0,00
		2	1	6.3.3	0,85
		5		Deflection	0,58
		5		Deflection	0,14
2	HEB160	2	1	6.2.4	0,19
		2	1	6.2.5	1,52
		2	1	6.2.6	0,21
		2	1	6.2.8	1,52
		2	1	6.2.9.1	1,52
		1	1	6.3.2.1	0,00
		2	1	6.3.3	1,15
		5		Deflection	0,79
		5		Deflection	0,42
3	HFRHS200X200X16	2	1	6.2.4	0,02
		2	1	6.2.5	1,01
		2	1	6.2.6	0,31
		2	1	6.2.8	1,01
		2	1	6.2.9.1	1,01
		2	1	6.3.3	1,05
		5		Deflection	1,38
		5		Deflection	0,57

2.6 CALCULATION OF UNITY CHECKS

2.6.1 Beam 1 - HEB160 (S 235)

Compression

art. 6.2.4

Combination: 2 x = 0 mm Nx = -226,027 kN Vz = -35,136 kN My = 0 kNm

$$N_{c,Rd} = \frac{A f_y}{\gamma_{M0}} = \frac{5427,5 \times 235}{1,00} \times 10^{-3} = 1275,472 \text{ kN} \quad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{226,0}{1275,5} = 0,18 < 1,0 \quad (6.9)$$

Bending moment (decisive)

art. 6.2.5

Combination: 2 x = 2500 mm Nx = -226,027 kN Vz = -35,136 kN My = -87,84 kNm

$$M_{y,c,Rd} = M_{pl,y} = \frac{W_{pl,y} f_y}{\gamma_{M0}} = \frac{354113 \times 235}{1,00} \times 10^{-6} = 83,217 \text{ kNm} \quad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{87,8}{83,2} = 1,06 > 1,0 \quad \text{not OK!} \quad (6.12)$$

Shear**art. 6.2.6**Combination: 2 $x = 0 \text{ mm}$ $N_x = -226,027 \text{ kN}$ $V_z = -35,136 \text{ kN}$ $M_y = 0 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{1762 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 239,1 \text{ kN} \quad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{35,1}{239,1} = 0,15 < 1,0 \quad (6.17)$$

Bending and shear**art. 6.2.8**Combination: 2 $x = 2500 \text{ mm}$ $N_x = -226,027 \text{ kN}$ $V_z = -35,136 \text{ kN}$ $M_y = -87,84 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{1762 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 239,1 \text{ kN} \quad (6.18)$$

$$V_{z,Ed} = 35,136 \text{ kN} < V_{z,pl,Rd} / 2 = 239,063 / 2 = 119,532 \text{ kN}$$

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force**art. 6.2.9**Combination: 2 $x = 2500 \text{ mm}$ $N_x = -226,027 \text{ kN}$ $V_z = -35,136 \text{ kN}$ $M_y = -87,84 \text{ kNm}$ **$M_{y,Ed} > M_{pl,y,Rd} = 83,217 \text{ kNm}$ not OK!****Lateral-torsional buckling resistance****art. 6.3.2.1**Combination: 1 $x = 2500 \text{ mm}$ $N_x = -197,946 \text{ kN}$ $V_z = -31,922 \text{ kN}$ $M_y = -79,806 \text{ kNm}$

$$\text{Number of lateral restraints: } 0 \quad (d')^2 b^3 t = 147^2 \times 160^3 \times 13,0$$

$$d' = h - t = 160 - 13 = 147 \text{ mm} \quad I_w = \frac{24}{24} = \frac{24}{24} = 48 \times 10^9 \text{ mm}^6$$

$$\text{torsional stiffness according to Roark case 26} \quad I_t = 313664 \text{ mm}^4$$

according to NEN-EN 1993-1-1+C2+A1/NB:2016 nl figure NB.33 and NB.34:

$$L_g = 2500 \text{ mm} \quad L_{st} = 2500 \text{ mm}$$

$$M_{y,1,Ed} = 0 \text{ kNm} \quad M_{y,2,Ed} = -79,806 \text{ kNm} \quad M_{y,Ed} (x=L_{st}/2 = 1250 \text{ mm}) = -39,903 \text{ kNm}$$

Calculated equivalent load $q = 0 \text{ kN/m}$

$$B^* = \frac{8 M}{8 |M| + q L_{st}^2} = \frac{8 \times -79,806 \times 10^6}{8 \times |-79,806 \times 10^6| + 0 \times 2500^2} = -1 \quad \text{D.4.3 (3)}$$

$$\beta = \frac{M_{y,1,Ed}}{M_{y,2,Ed}} = \frac{0}{-79,806} = 0 \quad C_1 = 1,803 \quad C_2 = 0$$

aangrijpingspunt belasting op $z = 0 \text{ mm}$

$$L_{kip} = L_{st} = 2500 \text{ mm}$$

$$S = \frac{h}{2} \times \sqrt{\frac{E \times I_z}{G \times I_t}} = \frac{160}{2} \times \sqrt{\frac{210000 \times 8892613}{80769 \times 313664}} = 687 \text{ mm} \quad (\text{NB.159})$$

$$C = \frac{\pi \times C_1 \times L_g}{L_{kip}} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times S^2}{L_{kip}^2} \times (C_2^2 + 1) \right)} + \frac{\pi \times C_2 \times S}{L_{kip}} \right) =$$

$$= \frac{\pi \times 1,803 \times 2500}{2500} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times 687^2}{2500^2} \times (0^2 + 1) \right)} + \frac{\pi \times 0 \times 687}{2500} \right) = 7,481 \quad (\text{NB.157})$$

$$h/t_w = 160/8 = 20 < 75 \quad \rightarrow k_{red} = 1 \quad (\text{NB.153})$$

$$M_{cr} = k_{red} \times \frac{C}{L_g} \times \sqrt{E \times I_z \times G \times I_t} =$$

$$= 1 \times \frac{7,481}{2500} \times \sqrt{210000 \times 8892613 \times 80769 \times 313664} \times 10^{-6} = 650,886 \text{ kNm} \quad (\text{NB.148})$$

$$\lambda_{Lt} = \sqrt{\frac{W_y f_y}{M_{cr}}} = \sqrt{\frac{354113 \times 235}{650886230}} = 0,358 < \lambda_{Lt,0} = 0,4 \quad \rightarrow \chi_{Lt} = 1,00$$

$$\lambda_{Lt} = 0,358 < \lambda_{Lt,0} = 0,4 \rightarrow \chi_{Lt} = 1,00$$

Uniform members in bending and axial compression

art. 6.3.3

Combination: 2 x = 2500 mm Nx = -226,027 kN Vz = -35,136 kN My = -87,839 kNm

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_y = \frac{L_{cr,y}}{i_y} \frac{1}{\lambda_1} = \frac{2500}{67,8} \frac{1}{93,9} = 0,393 \quad (6.50)$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_z = \frac{L_{cr,z}}{i_z} \frac{1}{\lambda_1} = \frac{2500}{40,5} \frac{1}{93,9} = 0,658 \quad (6.50)$$

Buckling curve $y-y$ b $\alpha = 0,34$

$$\Phi_y = 0,5 [1 + \alpha (\lambda_y - 0,2) + \lambda_y^2] = 0,5 \times [1 + 0,34 \times (0,393 - 0,2) + 0,393^2] = 0,61$$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}} = \frac{1}{0,61 + \sqrt{0,61^2 - 0,393^2}} = 0,929 \quad (6.49)$$

Buckling curve $z-z$ c $\alpha = 0,49$

$$\Phi_z = 0,5 [1 + \alpha (\lambda_z - 0,2) + \lambda_z^2] = 0,5 \times [1 + 0,49 \times (0,658 - 0,2) + 0,658^2] = 0,828$$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{0,828 + \sqrt{0,828^2 - 0,658^2}} = 0,751 \quad (6.49)$$

$$N_{Rk} = f_y A = 235 \times 5428 \times 10^{-3} = 1275,5 \text{ kN}$$

$$M_{y,Rk} = f_y W_{pl,y} = 235 \times 354113 \times 10^{-6} = 83,2 \text{ kNm}$$

$$M_{z,Rk} = f_y W_{pl,z} = 235 \times 169986 \times 10^{-6} = 39,9 \text{ kNm}$$

Interaction factors according to alternative method 2 (EN 1993-1-1, Annex B)

$$\varphi = M_2 / M_1 = 0 / -87,839 = 0 \quad \rightarrow C_{my} = 0,6 + 0,4 \varphi = 0,6 + 0,4 \times 0 = 0,6 > 0,4$$

$$k_{yy} = C_{my} \left(1 + (\lambda_y - 0,2) \frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}} \right) = 0,6 \times \left(1 + (0,393 - 0,2) \times \frac{226,027}{0,929 \times 1275,472 / 1,00} \right) = 0,622$$

$$k_{zy} = 0$$

$$\frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}} + k_{yy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} \frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{226,027}{0,929 \times 1275,472} + 0,622 \times \frac{87,839}{1 \times \frac{83,217}{1,00}} = 0,85 < 1 \quad (6.61)$$

$$\frac{N_{Ed}}{\chi_z N_{Rk} / \gamma_{M1}} + k_{zy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} \frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{226,027}{0,751 \times 1275,472} + 0 \times \frac{87,839}{1 \times \frac{83,217}{1,00}} = 0,24 < 1 \quad (6.62)$$

Deflection

$$\text{Combination: 5} \quad x = 1433 \text{ mm} \quad N_x = -194,331 \text{ kN} \quad V_z = -30,121 \text{ kN} \quad M_y = -43,165 \text{ kNm}$$

$$\text{Local node displacements } d_{z1} = 0 \text{ mm} \quad d_{z2} = -8 \text{ mm}$$

$$w_{fin.,z} = w_z - w_{Pre-camber,z} = 5,8 - 0 = 5,8 \text{ mm}$$

$$\frac{|w_{fin.,z}|}{w_{fin.,z,max}} = \frac{|5,8|}{2500 / 250} = \frac{|5,8|}{10} = 0,58 < 1,0$$

$$w_{add.,z} = w_z - w_{SLS Permanent,z} = 5,8 - 4,7 = 1 \text{ mm}$$

$$\frac{|w_{add.,z}|}{w_{add.,z,max}} = \frac{|1|}{2500 / 333} = \frac{|1|}{7,5} = 0,14 < 1,0$$

2.6.2 Beam 2 - HEB160 (S 235)

Compression

art. 6.2.4

$$\text{Combination: 2} \quad x = 0 \text{ mm} \quad N_x = -241,496 \text{ kN} \quad V_z = 50,669 \text{ kN} \quad M_y = 0 \text{ kNm}$$

$$N_{c,Rd} = \frac{A f_y}{\gamma_{M0}} = \frac{5427,5 \times 235}{1,00} \times 10^{-3} = 1275,472 \text{ kN} \quad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{241,5}{1275,5} = 0,19 < 1,0 \quad (6.9)$$

Bending moment (decisive)

art. 6.2.5

Combination: 2 $x = 2500 \text{ mm}$ $N_x = -241,496 \text{ kN}$ $V_z = 50,669 \text{ kN}$ $M_y = 126,672 \text{ kNm}$

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y} f_y}{\gamma_{M0}} = \frac{354113 \times 235}{1,00} \times 10^{-6} = 83,217 \text{ kNm} \quad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{126,7}{83,2} = 1,52 > 1,0 \quad \text{not OK!} \quad (6.12)$$

Shear

art. 6.2.6

Combination: 2 $x = 0 \text{ mm}$ $N_x = -241,496 \text{ kN}$ $V_z = 50,669 \text{ kN}$ $M_y = 0 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{1762 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 239,1 \text{ kN} \quad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{50,7}{239,1} = 0,21 < 1,0 \quad (6.17)$$

Bending and shear

art. 6.2.8

Combination: 2 $x = 2500 \text{ mm}$ $N_x = -241,496 \text{ kN}$ $V_z = 50,669 \text{ kN}$ $M_y = 126,672 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{1762 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 239,1 \text{ kN} \quad (6.18)$$

$$V_{z,Ed} = 50,669 \text{ kN} < V_{z,pl,Rd} / 2 = 239,063 / 2 = 119,532 \text{ kN}$$

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force

art. 6.2.9

Combination: 2 $x = 2500 \text{ mm}$ $N_x = -241,496 \text{ kN}$ $V_z = 50,669 \text{ kN}$ $M_y = 126,672 \text{ kNm}$

$$M_{y,Ed} > M_{pl,y,Rd} = 83,217 \text{ kNm} \quad \text{not OK!}$$

Lateral-torsional buckling resistance

art. 6.3.2.1

Combination: 1 $x = 2500 \text{ mm}$ $N_x = -205,954 \text{ kN}$ $V_z = 40,291 \text{ kN}$ $M_y = 100,727 \text{ kNm}$

$$\text{Number of lateral restraints: } 0 \quad (d')^2 b^3 t = \frac{147^2 \times 160^3 \times 13,0}{24} = 48 \times 10^9 \text{ mm}^6$$

$$d' = h - t = 160 - 13 = 147 \text{ mm} \quad I_w = \frac{147^2 \times 160^3 \times 13,0}{24} = 48 \times 10^9 \text{ mm}^6$$

torsional stiffness according to Roark case 26

$$I_t = 313664 \text{ mm}^4$$

according to NEN-EN 1993-1-1+C2+A1/NB:2016 nl figure NB.33 and NB.34:

$$L_g = 2500 \text{ mm} \quad L_{st} = 2500 \text{ mm}$$

$$M_{y,1,Ed} = 0 \text{ kNm} \quad M_{y,2,Ed} = 100,727 \text{ kNm} \quad M_{yEd} (x=L_{st}/2 = 1250 \text{ mm}) = 50,364 \text{ kNm}$$

Calculated equivalent load $q = 0 \text{ kN/m}$

$$B^* = \frac{8 M}{8 |M| + q L_{st}^2} = \frac{8 \times 100,727 \times 10^6}{8 \times |100,727 \times 10^6| + 0 \times 2500^2} = 1 \quad \text{D.4.3 (3)}$$

$$\beta = \frac{M_{y,1,Ed}}{M_{y,2,Ed}} = \frac{0}{100,727} = 0 \quad C_1 = 1,803 \quad C_2 = 0$$

aangrijpingspunt belasting op $z = 0 \text{ mm}$

$$L_{kip} = L_{st} = 2500 \text{ mm}$$

$$S = \frac{h}{2} \times \sqrt{\frac{E \times I_z}{G \times I_t}} = \frac{160}{2} \times \sqrt{\frac{210000 \times 8892613}{80769 \times 313664}} = 687 \text{ mm} \quad \text{(NB.159)}$$

$$C = \frac{\pi \times C_1 \times L_g}{L_{kip}} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times S^2}{L_{kip}^2} \times (C_2^2 + 1) \right)} + \frac{\pi \times C_2 \times S}{L_{kip}} \right) =$$

$$= \frac{\pi \times 1,803 \times 2500}{2500} \times \left(\sqrt{1 + \left(\frac{\pi^2 \times 687^2}{2500^2} \times (0^2 + 1) \right)} + \frac{\pi \times 0 \times 687}{2500} \right) = 7,481 \quad \text{(NB.157)}$$

$$h/t_w = 160/8 = 20 < 75 \quad \rightarrow k_{red} = 1 \quad \text{(NB.153)}$$

$$M_{cr} = k_{red} \times \frac{C}{L_g} \times \sqrt{E \times I_z \times G \times I_t} =$$

$$= 1 \times \frac{7,481}{2500} \times \sqrt{210000 \times 8892613 \times 80769 \times 313664} \times 10^{-6} = 650,886 \text{ kNm} \quad \text{(NB.148)}$$

$$\lambda_{Lt} = \sqrt{\frac{W_y f_y}{M_{cr}}} = \sqrt{\frac{354113 \times 235}{650886140}} = 0,358 < \lambda_{Lt,0} = 0,4 \quad \rightarrow \chi_{Lt} = 1,00$$

$$\lambda_{Lt} = 0,358 < \lambda_{Lt,0} = 0,4 \rightarrow \chi_{Lt} = 1,00$$

Uniform members in bending and axial compression

art. 6.3.3

Combination: 2 $x = 2500 \text{ mm}$ $N_x = -241,496 \text{ kN}$ $V_z = 50,669 \text{ kN}$ $M_y = 126,672 \text{ kNm}$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_y = \frac{L_{cr,y}}{i_y} \frac{1}{\lambda_1} = \frac{2500}{67,8} \frac{1}{93,9} = 0,393 \quad \text{(6.50)}$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_z = \frac{L_{cr,z}}{i_z} \frac{1}{\lambda_1} = \frac{2500}{40,5} \frac{1}{93,9} = 0,658 \quad \text{(6.50)}$$

$$\text{Buckling curve } y-y \text{ b} \quad \alpha = 0,34$$

$$\Phi_y = 0,5 [1 + \alpha (\lambda_y - 0,2) + \lambda_y^2] = 0,5 \times [1 + 0,34 \times (0,393 - 0,2) + 0,393^2] = 0,61$$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}} = \frac{1}{0,61 + \sqrt{0,61^2 - 0,393^2}} = 0,929 \quad (6.49)$$

Buckling curve $z-z$ $\alpha = 0,49$

$$\Phi_z = 0,5 [1 + \alpha (\lambda_z - 0,2) + \lambda_z^2] = 0,5 \times [1 + 0,49 \times (0,658 - 0,2) + 0,658^2] = 0,828$$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{0,828 + \sqrt{0,828^2 - 0,658^2}} = 0,751 \quad (6.49)$$

$$N_{Rk} = f_y A = 235 \times 5428 \times 10^{-3} = 1275,5 \text{ kN}$$

$$M_{y,Rk} = f_y W_{pl,y} = 235 \times 354113 \times 10^{-6} = 83,2 \text{ kNm}$$

$$M_{z,Rk} = f_y W_{pl,z} = 235 \times 169986 \times 10^{-6} = 39,9 \text{ kNm}$$

Interaction factors according to alternative method 2 (EN 1993-1-1, Annex B)

$$\varphi = M_z / M_1 = 0/126,672 = 0 \rightarrow C_{my} = 0,6 + 0,4 \varphi = 0,6 + 0,4 \times 0 = 0,6 > 0,4$$

$$k_{yy} = C_{my} \left(1 + (\lambda_y - 0,2) \frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}} \right) = 0,6 \times \left(1 + (0,393 - 0,2) \times \frac{241,496}{0,929 \times 1275,472 / 1,00} \right) = 0,624$$

$$k_{zy} = 0$$

$$\frac{\frac{N_{Ed}}{\chi_y N_{Rk}}}{\gamma_{M1}} + k_{yy} \frac{\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} M_{y,Rk}}}{\gamma_{M1}} = \frac{241,496}{0,929 \times 1275,472} + 0,624 \times \frac{126,672}{1 \times 83,217} = 1,15 > 1 \text{ not OK!} \quad (6.61)$$

$$\frac{\frac{N_{Ed}}{\chi_z N_{Rk}}}{\gamma_{M1}} + k_{zy} \frac{\frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} M_{y,Rk}}}{\gamma_{M1}} = \frac{241,496}{0,751 \times 1275,472} + 0 \times \frac{126,672}{1 \times 83,217} = 0,25 < 1 \quad (6.62)$$

Deflection

$$\text{Combination: 5} \quad x = 1433 \text{ mm} \quad N_x = -205,597 \text{ kN} \quad V_z = 41,387 \text{ kN} \quad M_y = 59,293 \text{ kNm}$$

$$\text{Local node displacements } d_{z1} = 0 \text{ mm} \quad d_{z2} = -7,9 \text{ mm}$$

$$w_{fin,z} = w_z - w_{\text{Pre-camber},z} = -7,9 - 0 = -7,9 \text{ mm}$$

$$\frac{|w_{fin,z}|}{w_{fin,z,max}} = \frac{|-7,9|}{2500 / 250} = \frac{|-7,9|}{10} = 0,79 < 1,0$$

$$W_{add,z} = W_z - W_{SLS\ Permanent,z} = -7,9 + 4,7 = -3,2 \text{ mm}$$

$$\frac{|W_{add,z}|}{W_{add,z,max}} = \frac{|-3,2|}{2500 / 333} = \frac{|-3,2|}{7,5} = 0,42 < 1,0$$

2.6.3 Beam 3 - HFRHS200X200X16 (S 235)

Compression

art. 6.2.4

Combination: 2 $x = 2431 \text{ mm}$ $N_x = -48,329 \text{ kN}$ $V_z = 0 \text{ kN}$ $M_y = 187,1 \text{ kNm}$

$$N_{c,Rd} = \frac{A f_y}{\gamma_{M0}} = \frac{11501,3 \times 235}{1,00} \times 10^{-3} = 2702,808 \text{ kN} \quad (6.10)$$

$$\frac{N_{Ed}}{N_{c,Rd}} = \frac{48,3}{2702,8} = 0,02 < 1,0 \quad (6.9)$$

Bending moment

art. 6.2.5

Combination: 2 $x = 2402 \text{ mm}$ $N_x = -48,228 \text{ kN}$ $V_z = 0 \text{ kN}$ $M_y = 187,327 \text{ kNm}$

$$M_{y,c,Rd} = M_{pl,y,Rd} = \frac{W_{pl,y} f_y}{\gamma_{M0}} = \frac{785442 \times 235}{1,00} \times 10^{-6} = 184,579 \text{ kNm} \quad (6.13)$$

$$\frac{M_{y,Ed}}{M_{y,c,Rd}} = \frac{187,3}{184,6} = 1,01 > 1,0 \quad \text{not OK!} \quad (6.12)$$

Shear

art. 6.2.6

Combination: 2 $x = 5000 \text{ mm}$ $N_x = -48,228 \text{ kN}$ $V_z = -241,739 \text{ kN}$ $M_y = -126,675 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{5751 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 780,2 \text{ kN} \quad (6.18)$$

$$\frac{V_{z,Ed}}{V_{c,z,Rd}} = \frac{241,7}{780,2} = 0,31 < 1,0 \quad (6.17)$$

Bending and shear

art. 6.2.8

Combination: 2 $x = 2402 \text{ mm}$ $N_x = -48,228 \text{ kN}$ $V_z = 0 \text{ kN}$ $M_y = 187,327 \text{ kNm}$

$$V_{c,z,Rd} = V_{pl,z,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}} = \frac{5751 \times (235 / \sqrt{3})}{1,00} \times 10^{-3} = 780,2 \text{ kN} \quad (6.18)$$

$$V_{z,Ed} = 0,000 \text{ kN} < V_{z,pl,Rd} / 2 = 780,212 / 2 = 390,106 \text{ kN}$$

The effect of the shear force on the moment resistance can be neglected. (2)

Bending and axial force

art. 6.2.9

Combination: 2 $x = 2402 \text{ mm}$ $N_x = -48,228 \text{ kN}$ $V_z = 0 \text{ kN}$ $M_y = 187,327 \text{ kNm}$ $M_{y,Ed} > M_{pl,y,Rd} = 184,579 \text{ kNm}$ **not OK!****Uniform members in bending and axial compression**

art. 6.3.3

Combination: 2 $x = 2402 \text{ mm}$ $N_x = -48,228 \text{ kN}$ $V_z = 145,984 \text{ kN}$ $M_y = 187,327 \text{ kNm}$

$$\lambda_{y1} = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_y = \frac{L_{cr,y}}{i_y} \frac{1}{\lambda_{y1}} = \frac{5000}{74,6} \frac{1}{93,9} = 0,714 \quad (6.50)$$

$$\lambda_{z1} = \pi \sqrt{\frac{E}{f_y}} = \pi \sqrt{\frac{210000}{235}} = 93,9 \quad \lambda_z = \frac{L_{cr,z}}{i_z} \frac{1}{\lambda_{z1}} = \frac{1666,7}{74,6} \frac{1}{93,9} = 0,238 \quad (6.50)$$

Buckling curve $y-y$ $\alpha = 0,21$

$$\Phi_y = 0,5 [1 + \alpha (\lambda_y - 0,2) + \lambda_y^2] = 0,5 \times [1 + 0,21 \times (0,714 - 0,2) + 0,714^2] = 0,809$$

$$\chi_y = \frac{1}{\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}} = \frac{1}{0,809 + \sqrt{0,809^2 - 0,714^2}} = 0,841 \quad (6.49)$$

Buckling curve $z-z$ $\alpha = 0,21$

$$\Phi_z = 0,5 [1 + \alpha (\lambda_z - 0,2) + \lambda_z^2] = 0,5 \times [1 + 0,21 \times (0,238 - 0,2) + 0,238^2] = 0,532$$

$$\chi_z = \frac{1}{\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}} = \frac{1}{0,532 + \sqrt{0,532^2 - 0,238^2}} = 0,992 \quad (6.49)$$

$$N_{Rk} = f_y A = 235 \times 11501 \times 10^{-3} = 2702,8 \text{ kN}$$

$$M_{y,Rk} = f_y W_{pl,y} = 235 \times 785442 \times 10^{-6} = 184,6 \text{ kNm}$$

$$M_{z,Rk} = f_y W_{pl,z} = 235 \times 785442 \times 10^{-6} = 184,6 \text{ kNm}$$

Interaction factors according to alternative method 2 (EN 1993-1-1, Annex B)

$$\varphi = M_2 / M_1 = -81,144 / 162,159 = -0,5 \quad \alpha_h = M_h / M_s = 162,159 / 72,817 = 1$$

$$C_{my} = 0,95 + 0,05 \alpha_h = 0,95 + 0,05 \times 1 = 1$$

$$k_{yy} = C_{my} \left(1 + (\lambda_y - 0,2) \frac{N_{Ed}}{\chi_y N_{Rk} / \gamma_{M1}} \right) = 1 \times \left(1 + (0,714 - 0,2) \times \frac{48,228}{0,841 \times 2702,808 / 1,00} \right) = 1,011$$

$$k_{zy} = 0$$

$$\frac{N_{Ed}}{\gamma_{M1} N_{Rk}} + k_{yy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} \frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{48,228}{1,00 \times 2702,808} + 1,011 \times \frac{187,327}{1 \times 184,579} = 1,05 > 1 \text{ not OK!} \quad (6.61)$$

$$\frac{N_{Ed}}{\chi_z N_{Rk}} + k_{zy} \frac{M_{y,Ed} + \Delta M_{y,Ed}}{\chi_{Lt} \frac{M_{y,Rk}}{\gamma_{M1}}} = \frac{48,228}{0,992 \times 2702,808} + 0 \times \frac{187,327}{1 \times \frac{184,579}{1,00}} = 0,02 < 1 \quad (6.62)$$

Deflection

Combination: 5 x = 2684 mm Nx = -40,739 kN Vz = -20,312 kN My = 158,19 kNm

Local node displacements d_{z1} = -0,4 mm d_{z2} = -0,5 mm

$$W_{fin,z} = W_z - W_{Pre-camber,z} = -27,7 - 0 = -27,7 \text{ mm}$$

$$\frac{|W_{fin,z}|}{W_{fin,z,max}} = \frac{|-27,7|}{5000 / 250} = \frac{|-27,7|}{20} = 1,38 > 1,0 \quad \text{Exceeding!}$$

$$W_{add,z} = W_z - W_{SLS Permanent,z} = -27,7 + 19,1 = -8,6 \text{ mm}$$

$$\frac{|W_{add,z}|}{W_{add,z,max}} = \frac{|-8,6|}{5000 / 333} = \frac{|-8,6|}{15} = 0,57 < 1,0$$